Artificial surf reefs: a review of built projects, surf quality, sediment transport and durability, with implications for the Israeli coast

Coastal Engineering course paper

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Abstract

Artificial surf reefs are promoted as structures that create surfable waves, protect beaches and enhance habitat. This review compiles the 13 surf reefs built or reshaped since 1997 with traced plan shapes, nine three-dimensional reconstructions and traceable sources. Coastal-engineering theory (shoaling, breaker index, Iribarren number, breaking intensity, peel angle) applied to the reconstructed profiles shows that the crest depth sets which waves break and the ramp gradient how: over similar depth gaps a 1:17 ramp gives a plunging, surfable wave and a 1:3 rise a collapsing one. Reefs that work are large (25,000 m³ or more); the three that do not work are 4,900 m³ or less. No reef offers a beginner's wave. Where surveys can test it, the shoreline response is 2 to 6 % of standard salient predictions; on the Gold Coast it cannot be separated from nourishment. Six of seven geotextile or inflatable reefs suffered major damage within a few years, none of four rock reefs. In a three-criterion rating (surf quality; safety and longevity; sediment and storm protection), only Albany clearly delivers a high-quality wave, with safety problems; rock reefs rate best on durability, and the sediment benefit is unknown or confounded everywhere. At Haifa, Israel, the 0.4 m spring tide would make the wave depend on wave height, not on the tide; a rock reef with a crest 0.6 to 1.0 m below mean sea level, a 1:16 to 1:20 ramp and 15,000 to 25,000 m³ per 100 m is proposed for testing.

Keywords: artificial surf reef; multi-purpose reef; surf quality; breaking intensity; peel angle; submerged breakwater; shoreline response; geotextile sand containers; Israel; Haifa

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1 Introduction

1.1 Purpose of surf reefs

A surf reef is a submerged structure of rock or sand-filled geotextile containers, placed in or just outside the surf zone so that incoming waves break over it in a controlled way and produce a rideable wave. Three purposes are attached to such structures (Mead & Borrero, 2011). The first is surf amenity: a reef can create a break where the natural one is poor, or restore one lost to coastal works. The second is coastal protection: a submerged structure dissipates wave energy and, in theory, causes sand to accumulate in its lee as a salient or tombolo; Blacka et al. (2013) review such reefs as a "softer" alternative to seawalls with negligible visual intrusion. The third is ecological enhancement: hard, stable substrate supports more species than mobile sand, and colonisation of surf reefs begins within days (Mead & Borrero, 2011). Reefs that combine the three are called multi-purpose reefs (MPRs).

The record is mixed. Of 32 submerged constructed reefs reviewed for New South Wales, 29 were intended for coastal protection, and the resulting shoreline often differed significantly from the design prediction; eight were built to improve surfing, and approximately half of these were considered at least partially successful; most structures settled or scoured, so that the crest level differed from the design (Blacka et al., 2013, p. ii).

1.2 History in brief

Artificial reefs for habitat date back to 2000 BC on the west coast of India (Kurian, 1995, as cited in Mead & Borrero, 2011). A reef combining coastal protection, ecology and surfing was first proposed in a 1969 article in Surfer Magazine, and the first methods to quantify a surfing break were developed by J. R. Walker in the 1970s and 1980s (Mead & Borrero, 2011). The first reef counted in this review, Burkitts Reef at Bargara, Australia, is a natural platform reshaped by hand in 1997 (Howells & O'Neal, 2024); the first purpose-designed structure, Cables Reef in Perth, was built in 1999 (Mead & Borrero, 2011). Between 1999 and 2010 seven reefs were built in five countries, most from sand-filled geotextile containers (Mead & Borrero, 2011). From 2011 the projects shifted to quarried rock, and an inflatable, sand-ballasted reef (the Airwave) failed on installation in 2019 (Section 3).

1.3 Motivation and research questions

This review responds to two requirements of the course instructor: to give insight, computational where possible and following the theory of the course, into the effect of a reef on the waves and on surf quality for surfers of different levels; and to include the effect on sand transport and the combination of a surf reef with beach stabilisation. Published evaluations are mostly by the designers (for example Mead & Borrero, 2011) or cover single sites. The main independent synthesis (Blacka et al., 2013) concentrates on New South Wales and precedes the rock reefs of 2019 and 2025. No source found tabulates all built surf reefs with traceable geometry, cost, material and outcome, and none relates their geometry to surf quality.

The practical motivation is a planned surf reef at Haifa, on the Mediterranean coast of Israel, where conditions differ from those of the built reefs in three ways. The tide is very small, about 0.4 m at spring tide (Rosen, 2001), against 1.1 to 4.3 m at the reefs surveyed here (for the three south-west Australian reefs this is the predicted annual range, not a spring range; Section 3). Waves are mostly small, with significant heights H_s of 0.5 to 1.0 m the most common class (Spanier & Zviely, 2022; Bitan & Zviely, 2020). Sand moves strongly with the seasons; the author's unpublished field observation is a seasonal bed-level change of the order of 1 m at Haifa (Section 7). The review also tests eight observations of the author on built reefs, labelled (a) to (h): (a) reef volume, (b) sand transport, (c) soft materials, (d) the number of surfable days, (e) the failure mechanism of tubes, (f) depth gap and ramp length, (g) surfer level and (h) the Israeli setting. Section 8 states for each whether the evidence supports it and whether it is consistent with published work or new.

The research questions are:

1. Which surf reefs have been built, of what size, material and cost, and how did they perform and last (Section 3)?
2. How does a reef change the waves, and what surf quality results for surfers of different levels, as set by depth gap, ramp length, breaker type, peel angle and tide (Section 4)?
3. What effect have surf reefs had on sand transport and the shoreline, and can a surf reef be combined with beach stabilisation where beaches are nourished (Section 5)?
4. Which materials survive in the breaking zone, and why do soft structures fail (Section 6)?
5. What do the answers imply for a reef at Haifa (Section 7)?

Section 2 gives the methods, Section 8 the discussion and Section 9 the conclusions. Appendix A documents the method for each reef, Appendix B shows the three-dimensional reconstructions, Appendix C holds the full data table, and Appendix D covers the other structures named by Bar and Drimer (2023) and the Israeli geotube projects.

2 Methods

2.1 Sources and the traceability standard

For each reef, the published design and monitoring papers, official reports, survey data sets, georeferenced aerial and satellite images, press articles and reef-specific web pages were collected. Peer-reviewed and agency reports were preferred; designers' papers (for example Mead & Borrero, 2011) are treated as the account of an interested party; press, surfer media and web databases are used only for facts that nothing else gives, and are named as such. Every number in the data set carries the value and unit, a full citation, the place where it was read (page, table, figure or file), the method, and a status: *sourced* (read in a cited source), *derived* (computed in this review from sourced inputs, a traced plan shape or a reconstructed three-dimensional model), *assumed* (flagged and numbered in Appendix A) or *unknown* (searched for and not found; no value given). Confidence is rated high, medium or low by a fixed rubric (Section 2.3; Appendix A.0). Where a statement rests on a second-hand source, for example Hutt et al. (2001) as reproduced by Lewis et al. (2015), the text says so. Values from secondary compilations whose cited sources could not be fetched and confirmed were not used. Costs were converted to US dollars at the World Bank annual-average exchange rate of the price year and to 2024 prices with the US consumer price index (World Bank, 2026); both steps are derived and inherit the uncertainty of the price year.

2.2 The reef sample

The sample is the 13 surf reefs built (for Burkitts, reshaped) with surfing as a primary or joint purpose, including those that failed or were removed and the unverified Xala reef (Mexico, 2025-26). It was compiled from Mead and Borrero (2011), Blacka et al. (2013), the web database of Raised Water Research and press reports, and is not claimed to be exhaustive. Borth is included because its northern mound was shaped as a surf reef; its southern oval is a breakwater. Of the eleven projects named by Bar and Drimer (2023), three are in the sample; the other eight, and the Israeli geotube projects, were not built as surf reefs and are described in Appendix D.

2.3 Plan shape, three-dimensional reconstruction and datums

Each outline was traced on the best available evidence, in a fixed order of preference: a measured survey of the structure, then a drawing of the built version, then a photograph. Satellite and aerial images (Esri World Imagery and dated Wayback releases; Esri et al., n.d.) were georeferenced from their tile bounds, drawings were scaled from a scale bar or a known dimension, and each outline was checked on a second image and against the dimensions in the text. Confidence is high if the structure is clearly visible on a georeferenced image or an as-built survey and the dimensions agree with the text to about 10 %, medium for a scaled design drawing, a partly visible structure or a 10-25 % disagreement, and low if the shape is inferred from text or a sketch. Outlines were obtained for nine reefs (Figure 1).

A three-dimensional model was built only where the plan, the crest level and the seabed depth were each sourced. The seabed is taken from the best measured data (soundings, lidar, multibeam read from a published figure, a coastal elevation model), otherwise from a chart (Garmin Navionics SonarChart, datum tested; Garmin Navionics, n.d.) or from design contours. The reef surface is the surveyed surface or a loft from the crest level over the traced toe outline down to the seabed; its volume is a mound envelope with voids included, not the fill volume. All heights are relative to mean sea level (MSL) through an explicit datum chain per reef (Appendix C, note 1), and only the tidal planes of the nearest standard port are used, for example those of Maritime Safety Queensland (2026) for the Gold Coast. Nine reefs were modelled (Appendix A, Table A1; Appendix B).

2.4 Depth profiles, depth gap and ramp length

The depth profile is the shore-normal section of the model through the shallowest crest cell. The crest depth C is the depth of the shallowest crest point below MSL (negative if the crest is above MSL); T is the seabed depth at the seaward toe of the traced outline; the depth gap is $G = T - C$. The ramp length L is the horizontal distance from the toe to the crest edge (the seaward-most cell within 0.25 m of the column maximum), and the mean slope G/L is quoted as $1{:}n$. Six reefs (Narrowneck, Cables, Albany, Palm Beach, Borth, Boscombe) have a ramp from audited geometry; the Pratte's ramp is assumed, and Mount Maunganui and Bunbury rest on hand estimates or a stand-in surface (Appendix C, note 2). Sourced crest depths are used where they exist (Cables, Palm Beach, Albany).

2.5 Computations

Sections 4 to 7 apply standard coastal-engineering relations to the profiles; formulas, thresholds and the status of every input are given where they are used. In outline: linear shoaling and Snell refraction for deep-water heights of 0.5 to 3 m and periods of 6 to 14 s at three tide levels, with breaking at $H = \gamma d$ ($\gamma = 0.78$), the breaker type from the surf-similarity (Iribarren) number (Battjes, 1974), the breaking intensity from the seabed gradient (Mead & Black, 2001b) and the peel angle against surfer skill (Hutt et al., 2001, as tabulated by Lewis et al., 2015) (Section 4); shoreline predictors (Black & Andrews, 2001; Ranasinghe & Turner, 2006) and wave transmission, compared with the surveyed seabed and shoreline change in the lee and at control sections, and longshore transport by the CERC formula (US Army Corps of Engineers, 2002) as an order of magnitude (Section 5); single-tube sliding and uplift after Bar and Drimer (2023) with a calibrated estimate, and rock stability by the Hudson formula (Section 6); and the same calculations with the Israeli tide (Section 7). Associations with outcome are Spearman rank correlations with permutation p values.

2.6 Limitations of the method

Data extraction was assisted by AI tools under the author's supervision; every value was traced to a cited source, and the author's decisions are listed in Appendix A. Readings from figures are by eye or by calibrated scales and carry the stated reading error. The original texts of Hutt et al. (2001), Mead and Black (2001a) and Black and Andrews (2001) were not read; their content is used as reproduced by later sources, and this is stated where it matters. No new field data were collected; the author's Haifa observations are used only in Section 7 and are labelled there. The consequences of the small sample, the unknown wave climates and the idealised seabeds are discussed in Section 8.2.

3 Survey of built surf reefs

3.1 The sample at a glance

Table 1 lists the 13 reefs in order of the start of construction; every value has its source, status and method in Appendix C. Seven reefs are made of geotextile or inflatable fabric filled with sand, five of rock (Burkitts is a natural platform reshaped) and one (Xala) of quarried boulders that is not yet documented. By the classes of this review, three reefs work for surfing (Burkitts, Palm Beach, Albany), three work rarely (Narrowneck, Mount Maunganui, Boscombe), two are mixed or disputed (Cables, Kovalam), three do not work (Pratte's, Opunake, Bunbury) and two are unknown (Borth, Xala). Figure 1 shows the plan shapes and depth profiles of the nine reefs with a traced outline, and Figure 2 the size, crest depth and depth gap of all 13.

Table 1. Master table of the 13 surf reefs. Volume as stated by the sources (rock or sand fill, not one common quantity; Borth derived from the survey edge; Opunake the design volume). Crest depth below MSL (negative: above MSL); depth gap = seabed depth at the seaward toe minus crest depth (Section 2.4). Cost: nominal local cost converted to USD at 2024 prices (derived; Section 2.1); Borth is the whole Phase 1 scheme. ? = unknown; † referred to chart datum (design stage). Outcome classes are readings of the cited sources by this review (Appendix C).

#	Reef	Country	Built	Purpose	Material	Volume (m^3)	Crest depth (m)	Depth gap (m)	Cost (USD M, 2024)	Outcome
1	Burkitts	AUS	1997	surf	natural basalt, boulders moved	300	?	?	0.015	works
2	Cables	AUS	1999	surf	granite boulders	5,500	1.55	4.96	2.19	mixed/disputed
3	Narrowneck	AUS	1999	multi	sand containers	70,000	2.32	3.81	2.79	works rarely
4	Pratte's	USA	2000	surf	sand bags (polypropylene)	1,350	2.68	1.35	1.00	does not work
5	Mount Maunganui	NZ	2005	multi	sand containers	2,800	2.03	2.70	1.54	works rarely
6	Opunake	NZ	2006	surf	sand bags	4,900	?	3.00†	1.55	does not work
7	Boscombe	UK	2008	multi	sand containers	12,800	0.90	4.56	7.29	works rarely
8	Kovalam	IND	2009	multi	sand containers	4,000	?	?	2.36	mixed/disputed
9	Borth	UK	2011	multi	quarry rock	30,119	-0.19	4.62	28.06	unknown
10	Palm Beach	AUS	2019	multi	quarry rock	25,000	1.50	5.17	15.52	works
11	Bunbury Airwave	AUS	2019	surf	inflatable bladder, sand ballast	105	0.68	2.24	0.49	does not work
12	Albany	AUS	2025	surf	quarry rock	34,000	1.00	6.33	7.57	works
13	Xala	MEX	2025	multi	quarry boulders	?	?	?	?	unknown

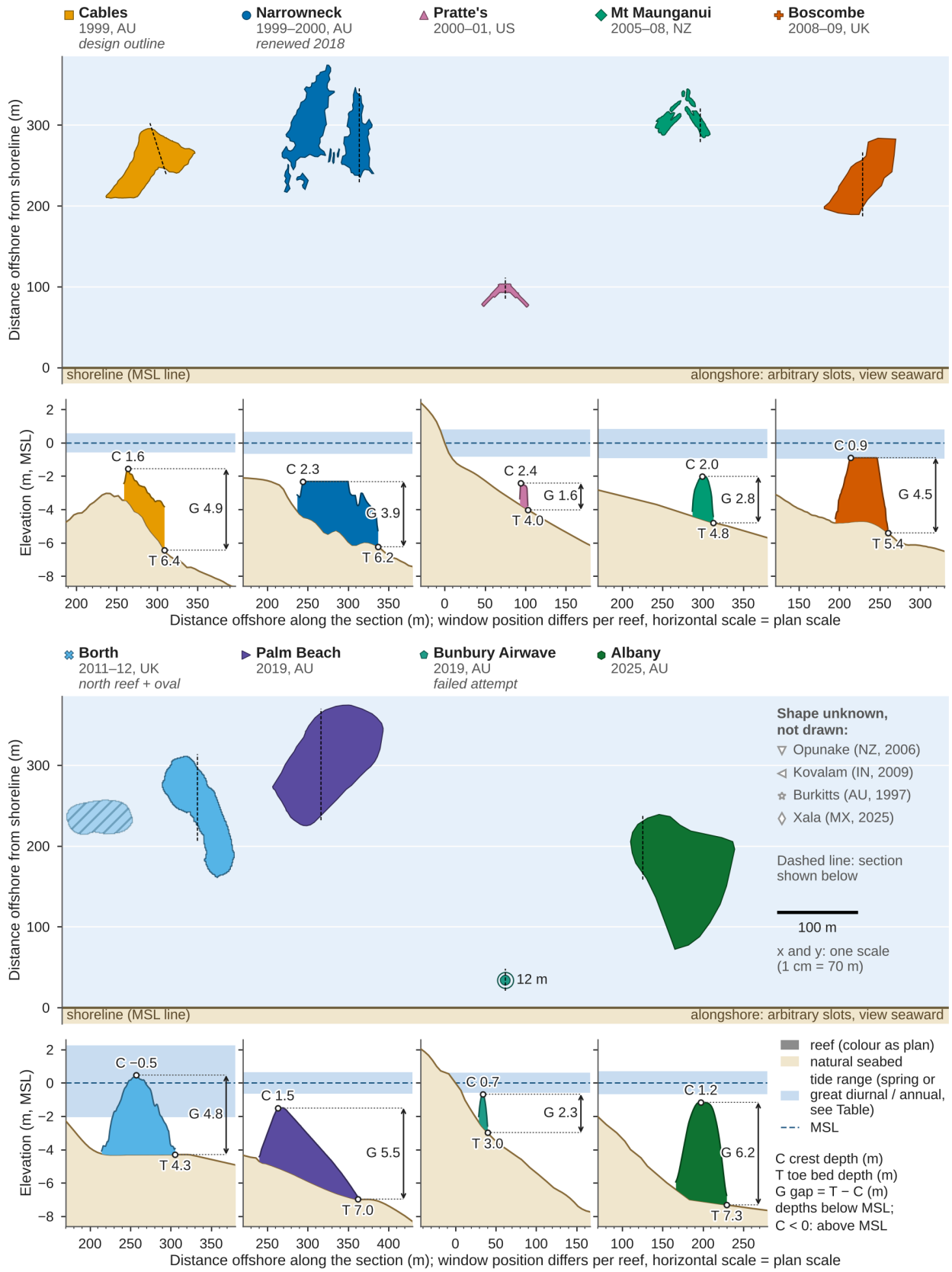


Figure 1. Plan shapes (upper part of each row) and cross-shore depth profiles (lower part) of the nine reefs with a traced outline, in order of construction (Opunake, Kovalam, Burkitts and Xala could not be traced). Plans share one metric scale (100 m bar), are viewed looking seaward and are set at their distance from the MSL shoreline (thick line); the hatched shape is Borth's southern oval, a breakwater. Each profile is the section of the 3D reconstruction along the dashed line, relative to MSL, with a vertical exaggeration of about 21:1; the blue band is the tide range (spring range; great diurnal range at Pratte's; predicted annual range of 2026 at Cables 1.14 m, Bunbury 1.20 m and Albany 1.38 m, Western Australia Department of Transport & Bureau of Meteorology, 2026), and C, T and G are the crest depth, toe depth and depth gap. Sources and methods: Appendix A.

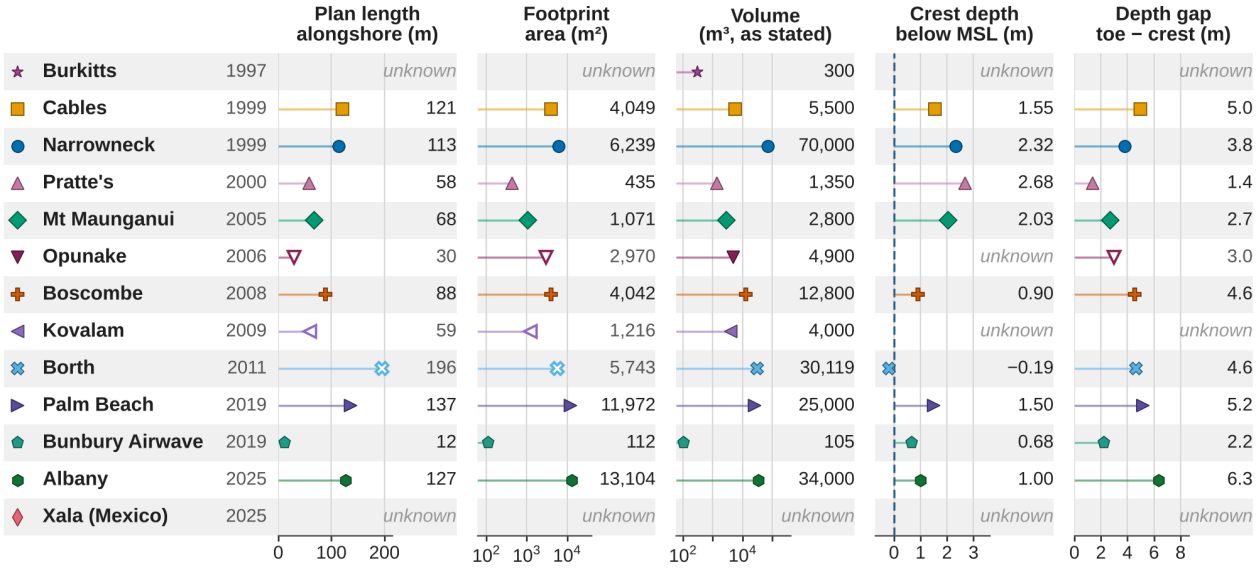


Figure 2. Size of the 13 projects in order of construction: alongshore length and footprint of the traced outline, volume as stated by the sources (rock or sand fill, not one common quantity), crest depth below MSL (negative: above MSL, Borth) and depth gap. Open symbols: assumed, low-confidence or not directly comparable values; 'unknown': no source checked gives the value. Lengths, areas, crest depths and gaps are derived from the outlines and 3D models; volumes are sourced except Borth and Bunbury (derived). Borth's plan area and volume come from different outlines: the area is the rock edge visible on the photograph of 17 September 2024, the volume (30,119 m^3) that of the 2022 laser-survey edge (8,675 m^2). Values and sources: Appendix C.

3.2 Australia

Cables Reef, Perth (1999). After a surfers' campaign from 1975, the Western Australian Department of Sport and Recreation financed a reef on the site of a natural break (Raised Water Research, 2019b). It was designed for swells of 0.5-3.5 m, a peel angle of about 45 degrees and a 1:20 ramp (Lyons, 1992; Pattiaratchi, 1999). Granite boulders of 1.5 t (body) and 3.0 t (perimeter) were placed on limestone in 1999 (Raised Water Research, 2019b) for AUD 1.8 M (Blacka et al., 2013). The reef is undamaged: surveyed rock volumes of 4,671 m^3 (1999) and 5,219 m^3 (2002) differ by less than the survey error. Its surf record is disputed: the designers report 142 surfable days a year against 5-7 before, counting days with surfers present (Mead & Borrero, 2011), whereas surfers report good waves only in big swells (Blacka et al., 2013, p. 93), and a 2009 critique puts the crest about 1.5 m below the lowest tide, in the swell shadow of Rottnest Island (Pitt, 2009, as cited in Raised Water Research, 2019b).

Narrowneck Reef, Gold Coast (1999-2000, renewed 2017-18). The primary aim was to protect the eroding Narrowneck isthmus, with surfing second (Jackson et al., 2007; Blacka et al., 2013). About 400 containers holding about 70,000 m^3 of sand were laid to a design crest of -1.0 m AHD tested on a 1:50 physical model (Mead & Borrero, 2011; Jackson et al., 2007), for AUD 2.3 M, or AUD 2.8 M with top-ups and maintenance (Blacka et al., 2013, p. 101). Non-uniform placement "detracted from the surfing amenity" (Mead & Borrero, 2011, p. 3); the containers settled by 1-2 m, the 2018 crest is at -2.2 m AHD, and 84 containers were renewed in 2017-18 (Blacka et al., 2013, pp. 97-98; Corbett et al., 2023). The reef "is rarely surfed" because world-class breaks work in the same conditions (EcoShape, n.d.). Its protective effect is disputed (Section 5.4).

Palm Beach Reef, Gold Coast (2019). The City of Gold Coast built the reef to protect the beach and to give a right-hand wave (Hunt et al., 2022), on a concept design by DHI (Mortensen et al., 2015). About 60,000 t of quarried boulders (25,000 m^3) were placed to a crest at -1.5 m MSL for AUD 18.2 M (Appendix C). The reef is stable, and a year of shore-camera tracking found the waves "desirable" when easterly or southerly swells had H_s above 0.8 m and T_p above 7 s, with rides longer than in the natural surf zone (Hunt et al., 2022; Prenzler et al., 2023).

Bunbury Airwave (2019). The Airwave, patented by T. Bottegal (Waveco), is a Hypalon bladder 12 m across ballasted with 140-150 t of beach sand, intended to turn a closeout into a peeling wave. The City of Bunbury invested AUD 75,000 (ABC News, 2019); the cost was about USD 350,000-400,000 (Raised Water Research, n.d.-a; sources disagree). The bladder tore along a glued seam during installation in December 2019 and was removed within about three days (ABC News, 2019); according to the founder, the seam opened a couple of days after installation and the wave "worked amazingly" until then (T. Bottegal, Airwave founder, pers. comm., 8 Oct 2026). The two accounts differ on timing and are both given.

Southern Ocean Surf Reef, Albany (2025). Bluecoast Consulting Engineers designed the reef from the feasibility study of Royal HaskoningDHV (2015) as a long peeling wave for surfing; the AUD 11.75 M cost was shared by the Australian Government (AUD 4.75 M), the Government of Western Australia (AUD 5 M), the City of Albany (AUD 1.995 M) and Albany Boardriders (City of Albany,

n.d.). Rock of three grades (62,800 t, design volume 34,000 m^3) was placed to a crest of -1.0 m AHD (City of Albany, n.d.). It opened in July 2025 with faces of 0.8-1.3 m and rides of 100-120 m (Raised Water Research, n.d.-d; whether the wave is the "Long Right" of the 2015 option or the left described at opening was not reconciled), but a designer calls it "intermediate-to-advanced on most days" (Mercer, 2025). A storm in May 2025 moved some core rock, and about 18 high rocks were repositioned in February 2026 after injuries (Sheil, 2026).

Burkitts Reef, Bargara (1997). G. Redgard, a sugarcane cutter, moved about 300 m^3 of boulders on a natural basalt platform with a hired excavator, at about AUD 10,000 of his own money and with no engineering design (Howells & O'Neal, 2024; Raised Water Research, n.d.-b). Surfable days rose from "a handful" to about 30 a year, a researcher's estimate (A. Pitt, as reported by Raised Water Research, n.d.-b), and the site has needed no maintenance in nearly three decades (Howells & O'Neal, 2024). No crest level, depth or plan is known.

3.3 New Zealand

Mount Maunganui Reef (2005-2008). Designed by K. Black and S. Mead as a symmetrical "delta" with a wave-focusing and a wave-breaking zone (Mead & Black, 1999; Mead et al., 2007), it was financed by a community trust at about NZD 1.5 M (SunLive, n.d.). Of 6,500 m^3 designed, 2,800 m^3 were built as 24 containers (Mead & Borrero, 2011, p. 3); containers ruptured during filling and the apex deflated to about -2.0 to -2.4 m CD by 2013 (Dahm & Gibberd, 2013). In October 2006 the right-hander broke "in a hollow, tubing manner", but after the crest slumped by about 1 m it "breaks during only a very confined set of conditions" (Mead & Borrero, 2011, p. 3). The reef was removed in 2014 (SunLive, n.d.).

Opunake Reef (2006-2008, never completed). Financed by the South Taranaki District Council and a local trust (Reid & Sumner, 2021), it was designed for a right-hand wave of 1.8-2.5 m face, to raise surfable days from about one in eight to one in three (Moores, 2006). Only 27 bags were in place in April 2008, costs approached twice the estimate (Mead & Borrero, 2011, p. 4; NZD 1.7 M, Appendix C), and the structure broke apart; the designers later called bag failure the "fatal flaw" of the technology (Reid & Sumner, 2021).

3.4 United Kingdom

Boscombe Surf Reef (2008-2009). Bournemouth Borough Council commissioned ASR Ltd, as part of a regeneration scheme, to improve a site with "mediocre" surfing; the reef was designed primarily for surfing (Mead et al., 2010). It cost GBP 3.2 M on a fixed-price contract (Mead & Borrero, 2011; Raised Water Research, n.d.-e) and aimed to double the good surfing days from about 77 to 154 a year (Raised Water Research, n.d.-e). Sand containers (12,800 m^3) were placed with a crest designed at +0.5 m CD to break small waves at low tide; whether it was built higher is unresolved (Rendle, 2016, p. 70; Mead et al., 2010). The independent evaluation found the wave "steep and frequently plunging" and restricted to more advanced surfers (Davidson, 2010, as cited in Rendle, 2016, pp. 70-71; Section 4.3). The containers failed from March-April 2011 and the reef was closed on 31 March 2011 (Rendle & Davidson, 2012).

Borth reef (2011-2012). The reef is part of the Phase 1 coastal defence of Borth, Wales, designed by Royal Haskoning with ASR as sub-consultant, physically modelled by HR Wallingford and funded by a Welsh Government grant and EU convergence funds (Borth Community, n.d.). The whole scheme cost GBP 13 M; about 51,000 t of rock lie in two mounds (30,119 m^3, derived), of which the northern is shaped as a surf reef (Appendix A.9). The crest is 0.19 m above MSL in a 4.3 m spring range, so it dries at low water. The defences withstood the storms of December 2011 and show no damage in imagery of 2012-2024 (Hansford, 2011; Daily Post, 2013; this review), but no surf record exists, and Blacka et al. (2013) rate the surfing objective "unknown".

3.5 United States, India and Mexico

Pratte's Reef, El Segundo (2000-2001). The first structure in US waters built for surfing, it mitigated the loss of surf caused by the El Segundo groin (Borrero & Nelsen, 2003) and was financed by Chevron as legal mitigation, with the California Coastal Conservancy and the Surfrider Foundation (Raised Water Research, n.d.-f). It was a V of 110 sand-filled polypropylene bags in Phase I and 90 in Phase II (1,350 m^3), costing USD 550,000 (Appendix C). Fewer than 25 people surfed it in two years, in about 10 sessions, because of its "small size": 30 m cross-shore against about 150 m at Narrowneck (Borrero & Nelsen, 2003, pp. 13-14). Anecdotal witness accounts (evidence grade D: viewer comments on the YouTube video *The $500k Surf Reef Failure*, Dan Harmon, 2026, retrieved 19 Oct 2026) agree: a worker at the adjacent power plant "never saw" a surfable wave, others describe it as breaking rarely, "really tide sensitive and wave height sensitive" and "barely surfable", and attribute this partly to shadowing of the southern swells by Catalina Island. The bags disintegrated within about two years, the crest dropped 1.5 m, and the reef was removed in 2008 and 2010 (Blacka et al., 2013; Fontaine, 2008); one commenter on the same video, by contrast, states that the bags he saw removed "were still intact" (Dan Harmon, 2026), so the failure is better documented for the wave than for the fabric.

Kovalam Reef, India (2009-2010). ASR built India's first multi-purpose reef from 5-6 fanned strips of sand containers (about 4,000 m^3; INR 75 M) for a left-hander of about 100 m, financed by the post-tsunami rehabilitation programme of the Government of India (Raised Water Research, n.d.-c; Mead & Borrero, 2011). The first season was praised as a wave that "could very well be the best wave an

artificial reef has ever created" (Collins, 2010), but later "most of the waves just roll over the reef" (Raised Water Research, n.d.-c); the containers were torn, deflated and sunk by 2013 (E. Rendle, pers. comm., 2013, in Blacka et al., 2013, p. 70). The designers' claim of a beach 10-30 m wider (Mead & Borrero, 2011) has not been independently assessed.

Xala Reef, Jalisco, Mexico (2025-26; unverified). Announced as part of a private resort development and built from quarried boulders ("3,000 truckloads"; Romeyn, 2025; Indelicato, 2026), the reef has no published crest level, volume, cost, surf report or monitoring. It carries no weight in the analysis.

3.6 What the survey shows

Four patterns emerge, to be tested in Sections 4 to 7. First, size separates the outcomes: the three reefs that do not work are 4,900 m^3 or less, whereas the two constructed reefs that work are 25,000 m^3 or more (Burkitts is a reshaped natural platform, not a built volume). Second, six of the seven fabric-filled or inflatable reefs suffered major damage or loss within a few years, whereas none of the four rock reefs failed (Appendix C, Table C4); material is confounded with date and budget. Third, reported surfable days are not comparable: they are a count of days with surfers present (Cables), a design baseline (Boscombe), a share of the year of unstated basis (Albany) or an estimate (Burkitts). Fourth, no reef was built or reported as a beginner wave. Design wave heights and periods are unknown for most reefs, and most performance claims come from the designers (Mead & Borrero, 2011).

3.7 Structures that are not surf reefs

Bar and Drimer (2023, p. 2) summarise the built record in one paragraph and borrow the statement that most projects "have failed" from the Raised Water Research website, without a list or a definition of failure. Of the eleven projects they name, three are among the 13 reefs above. Of the other eight, four are geotextile-tube shore-protection breakwaters (Korea, Mexico, the UAE, Indonesia), one is a nature-based breakwater (New York), two are unbuilt design studies (Portugal, Spain), and one is the Ashkelon North cliff-foot geotube scheme in Israel, where several tubes tore in the first winter (State Comptroller of Israel, 2020). None was built for surfing; they are documented in Appendix D.

4 Reef effect on waves and surf quality

A surf reef must make a swell break at a chosen place, with a chosen breaker shape, and with a break that peels along the reef at a speed surfers can follow. Section 4.1 summarises the theory; Sections 4.2 to 4.6 apply it to the nine reconstructed reefs (Section 2; Appendix A). Outcomes are scored 3 (works), 2 (works rarely or mixed) and 1 (does not work) from the survey (Appendix C).

4.1 Theory: shoaling, refraction, breaking, breaker type and peel

Shoaling. In linear (Airy) theory the wave number k follows from $\omega^2 = gk \tanh(kh)$, and conservation of energy flux, without refraction, friction or reflection, gives the shoaling coefficient

$$K_s = \frac{H}{H_0} = \frac{\sqrt{c_{g0}}}{c_g},$$

with H_0 the deep-water height and c_g the group velocity at depth h (Dean & Dalrymple, 1991). Over a reef the depth falls from toe to crest within tens of metres, so K_s rises quickly at the end of the wave's path.

Refraction and focusing. By Snell's law, $\sin\theta/c$ = const, oblique crests turn towards the depth contours, and convex contours make rays converge on a shoal (Dean & Dalrymple, 1991). Mead and Black (2001a) classified the seabed of 34 world-class breaks into ramp, platform, focus, wedge, ledge, ridge and pinnacle components, each of which pre-conditions the swell or sets the break point or breaker type (as summarised by Blacka et al., 2013, Section 2.6); a surf reef is designed to reproduce them.

Breaking criterion. A wave breaks when $H_b = \gamma h_b$. The flat-bed limit $\gamma = 0.78$ (McCowan, 1894, as given by Weggel, 1972) rises on slopes; Weggel (1972) gives γ_b as a function of bed slope and $H_b/(gT^2)$, and Battjes (1974) reports values of about 0.7 to 1.2. The criterion sets *which* waves break on a crest.

Breaker type. The surf-similarity (Iribarren) parameter at the break point,

$$\xi_b = \frac{\tan\beta}{\sqrt{H_b/L_0}}, \qquad L_0 = \frac{gT^2}{2\pi},$$

separates spilling ($\xi_b < 0.4$), plunging ($0.4 < \xi_b < 2.0$) and surging or collapsing breakers ($\xi_b > 2.0$) (Battjes, 1974, after Galvin, 1968). It sets *how* a wave breaks. For a 1 m breaker, $\xi_b = 0.4$ corresponds to a slope of 1:31 at $T = 10$ s and 1:22 at 7 s, and $\xi_b = 2.0$ to 1:6.2 and 1:4.4 (derived in this review).

Breaking intensity. Mead and Black (2001b) related the hollowness of plunging waves, measured by the vortex ratio Y (length to width of the tube), to the seabed gradient along the wave orthogonal, $Y = 0.065X + 0.821$ ($R^2 = 0.71$), where X is the gradient "given as the horizontal distance to one vertical unit", i.e. a slope of 1:X (Mead, 2000, Eq. 6.5, p. 115). Intensity is "extreme" for $Y = 1.6$ to 1.9, "very

high" for 1.91 to 2.2, "high" for 2.21 to 2.5, "medium/high" for 2.51 to 2.8 and "medium" for 2.81 to 3.1; waves with $Y < 1.6$ are likely to collapse and be unsurfable (Mead, 2000, Table 6.2, p. 121). The fit covers slopes of about 1:9 to 1:40 and scatters widely (Robertson et al., 2014, found $R^2 = 0.27$ on their field data), so it is used as an indicator only.

Peel angle and surfer skill. The peel angle α is the angle between the unbroken crest and the path of the break, which moves along the crest at the peel speed $V_p = c_b / \sin\alpha$, so small angles are fast (Lewis et al., 2015, p. 3). Hutt et al. (2001) gave the smallest peel angle that each surfer rating can ride: 90, 70, 60, 55, 50, 40, 29 and 27 degrees for ratings 1 (beginner) to 8 (professional), with no limit for ratings 9 and 10 (as reproduced by Lewis et al., 2015, Table 3, and Barlow, 2013, Table 1.1, which agree). Walker (1975, p. 5) described the 90-degree wave as suited to "a beginning surfer who has not developed skill levels sufficient to negotiate a turn" and suggested 30 degrees "as a minimum for a plunging breaker at 6-foot height": a hydrodynamic minimum, not a skill class. On a reef arm, Snell's law refracts the angle ϕ_0 between the incident crest and the arm at the toe (depth h_0) to $\sin\alpha_b = \sin\phi_0 \, c(h_b)/c(h_0)$ at the break (method of this review).

4.2 Where waves break: crest depth, depth gap and ramp length

Waves of $H_0 = 0.5$ to 3 m and $T = 6$ to 14 s were shoaled along the crest line of each reef at low, mean and high water (normal incidence, $\gamma = 0.78$; sensitivity with $\gamma = 0.55$ and the Weggel index). At MSL a wave needs H_0 of 0.3 to 1.3 m to break on the reef, and above 1.6 to 5.2 m it breaks seaward of it (Table 2; Borth, whose crest dries, excepted). Over all cases, 34 to 84 % break on the ramp or crest.

The author's observation (f) is that the depth gap alone does not decide the wave: a long, gradual ramp from 5 m to 2 m can give a good wave, because most waves do not break at 2 m, whereas a sudden rise to 1 to 1.5 m gives a too-hollow wave. Figure 3 tests this on two reefs with comparable depth gaps (5.49 m at Palm Beach, 4.51 m at Boscombe) under the same 1 m, 10 s wave. Both waves break at the same height and depth ($H_b = 1.39$ m, $h_b = 1.78$ m): the crest depth fixes where the wave breaks. But over Palm Beach's 89 m ramp (1:17) the breaker plunges with $\xi_b = 0.56$ and "very high" intensity ($Y = 2.05$), while over Boscombe's 13 m ramp (1:3.1) it collapses ($\xi_b = 3.00$, $Y = 1.05$, extrapolated).

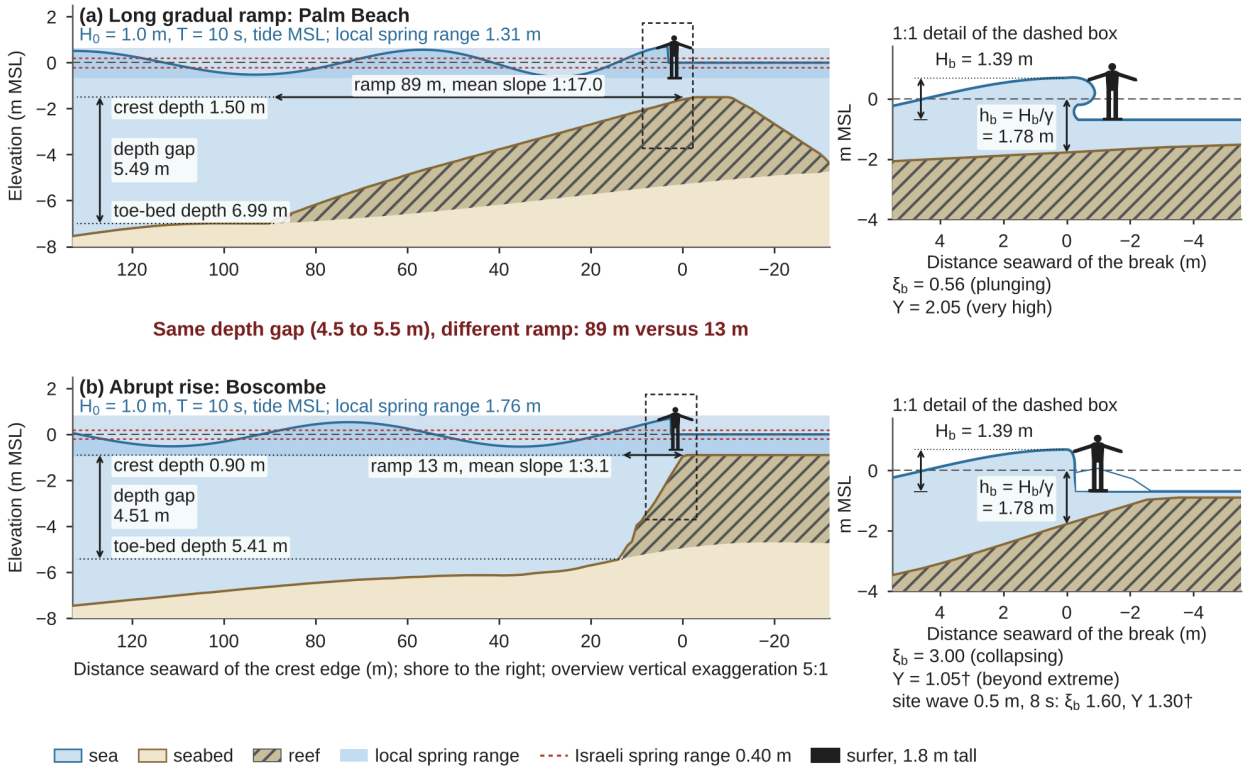


Figure 3. Shoaling and breaking over a long gradual ramp (a, Palm Beach) and an abrupt rise (b, Boscombe) along the crest line of each 3D model, for $H_0 = 1.0$ m, $T = 10$ s, normal incidence and MSL (linear shoaling, Dean & Dalrymple, 1991; breaking at $H = 0.78d$, McCowan, 1894, as given by Weggel, 1972). Overview panels are exaggerated 5:1; the dashed boxes are enlarged 1:1 with the breaker, the breaking depth $h_b = 1.78$ m ($H_b = 1.39$ m in both) and a 1.8 m standing surfer (lip and foam schematic). Similar depth gaps give a plunging breaker over the 89 m ramp (1:17; $\xi_b = 0.56$, Battjes, 1974; $Y = 2.05$, Mead & Black, 2001b) and a collapsing front over the 13 m ramp (1:3.1; $\xi_b = 3.00$, $Y = 1.05$, extrapolated beyond the 1:9 to 1:40 range of the fit). Blue band: local spring tide range; dashed red lines: the Israeli spring range of 0.40 m (Rosen, 2001).

Across the reefs (Figure 4), the depth gap ranks with the outcome (Spearman $\rho = 0.93$, permutation $p = 0.005$, $n = 8$), as does the ramp length ($\rho = 0.85$, $p = 0.015$), but not the crest depth ($\rho = 0.00$). The same 2 to 4 m gap gives a surfable wave over a long ramp and a collapsing wave over an abrupt rise. The ramp is a recognised seabed component (Mead & Black, 2001a, as summarised by Blacka et

al., 2013), but “there are no models available off the shelf which directly assess surfability” (Blacka et al., 2013, p. 56), and no source found quantifies the contrast between a gradual ramp and a sudden rise; the quantification is new to this review.

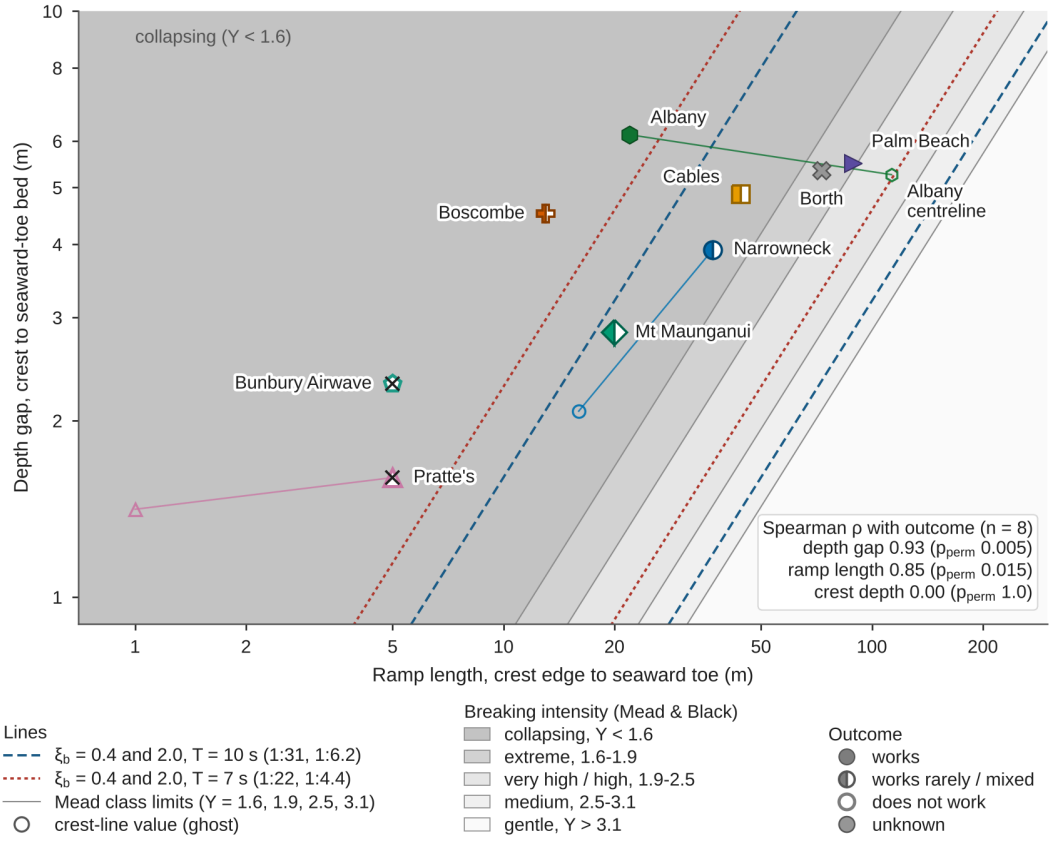


Figure 4. Depth gap against ramp length for the nine modelled reefs (logarithmic axes; lines of constant gradient 1:X are straight). Fill gives the outcome (filled: works; half: works rarely or mixed; open with a cross: does not work; grey: unknown); ghost symbols give alternative sections (Narrowneck second crest line, Pratte’s crest line, Albany centreline). Shading: breaking-intensity classes from $Y = 0.065X + 0.821$ (Mead & Black, 2001b; Mead, 2000); dashed and dotted lines: ξ_b = 0.4 and 2.0 (Battjes, 1974) for a 1 m breaker at T = 10 s and 7 s. The gradient includes the natural bed between crest and toe, so it can be steeper than the reef face (Cables 1:9 here, 1:20 on the face). Spearman ρ with outcome (n = 8): depth gap 0.93 (p = 0.005), ramp length 0.85 (0.015), crest depth 0.00 (1.0).

4.3 Breaker type, hollowness and the Boscombe case

Over all cases that break on the reef (Table 2), Palm Beach gives a median ξ_b of 0.49 and Y of 2.01 with no case below Y = 1.6, whereas Bunbury (1:2.7) and Boscombe (1:3) give Y < 1.6 in every case and Mount Maunganui (1:9) in 91 %. The median ξ_b ranks with the outcome at $\rho = -0.69$ (p = 0.078). Period matters as well as slope: the surging share rises from 0.21 at 6 s to 0.41 at 10 to 12 s, so the same reef gives gentler waves in short-period seas such as Israel’s. Hollowness alone is not failure: Mount Maunganui first “broke as designed in a hollow, tubing manner” (Mead & Borrero, 2011, p. 3), and Albany, whose crest line rises at about 1:4, works but is “intermediate-to-advanced on most days” (Mercer, 2025).

Boscombe tests the author’s observation (d) that its structure rose too steeply and made a wave only for bodyboarders. The computed 1:3 flank collapses in every case, Boscombe’s typical sea (0.5 m, 8 s) gives ξ_b = 1.60 and Y = 1.30, and even the 2011 survey of the damaged reef (mean ramp 1:12) gives Y < 1.6 in 85 % of cases. The independent audit found the wave “steep and frequently plunging”, so that “novice surfers and intermediates would not be able to surf the Boscombe Reef” (Davidson, 2010, as cited in Rendle, 2016, pp. 70-71); it required Hutt level 7 or 8 against a design level of 5 or 6 (Rendle, 2016, p. 205; the designers give 3 to 6, Mead et al., 2010). Bodyboarders were more satisfied than stand-up surfers, who called it “a bodyboarders wave” (Rendle, 2016, pp. 178, 204-205), and the designers note that outside the design swell the waves were “frequently more suitable to body-boarding due to the intense plunging nature of the wave breaking” (Mead et al., 2010, p. 5); yet advanced stand-up surfers rode it (Davidson, 2010, as cited in Rendle, 2016, p. 70). The evidence supports “too steep” and “mostly bodyboarders”, not “only bodyboarders”. The crest was set shallow so that the common 0.5 m waves would break at low tide (Rendle, 2016, p. 71), exactly the condition in which a short ramp collapses the wave, and, exposed at low tide and too deep at high tide, it left a mid-tide window of “1 to 2 hours” (Rendle, 2016, p. 71).

4.4 Peel angle and surfer skill

The designed arm angles of 45 to 81 degrees refract to 16 to 62 degrees at the break (Figure 5). Small waves break in shallow water, where refraction is strongest: at H_0 = 0.5 m every reef reaches α_b below 27 to 29 degrees (Hutt ratings 8 to 9), and only large waves keep α_b above 50 degrees. Peel speeds are 6.9 to 11.4 m/s. Only one case reaches the 60 to 90 degrees of ratings 1 to 3: Borth at H_0 = 3 m (61.9 degrees, rating 3), a large wave at the upper end of Borth's window, on a reef with no surf record; the incidence angle ϕ_0 is assumed for five of the eight reefs (Appendix A). The observations agree: no reef was built or reported as a beginner wave (Albany "intermediate-to-advanced", Mercer, 2025; Boscombe "an intermediate to expert break", Rendle, 2016, p. 273). The literature defines the beginner's wave by a large peel angle rather than by a spilling breaker; both are missing here, since spilling breakers are 0 to 29 % of on-reef cases, and a spilling ramp of about 1:30 needs 35,000 to 49,000 m³ per 100 m in Israeli conditions (Section 7).

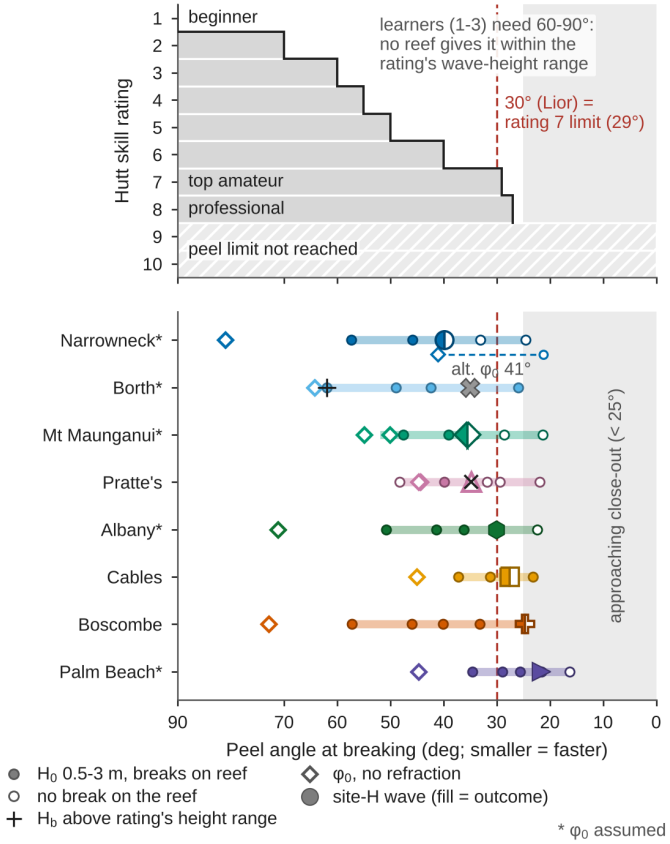
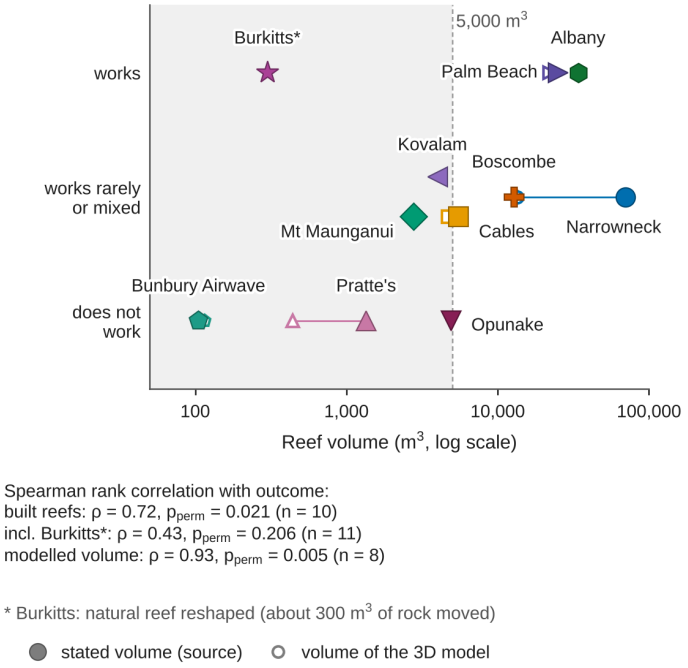


Figure 5. Peel angle at breaking and the surfer skill that can use it. Upper panel: the smallest peel angle per surfer rating after Hutt et al. (2001), as reproduced by Lewis et al. (2015) and Barlow (2013). Lower panel: for each reef the bar spans the refracted peel angle (Snell's law, no diffraction) at the site period for H_0 = 0.5 to 3 m, from the angle ϕ_0 between incident crest and reef arm (diamonds; assumed for five reefs, *); the large symbol is the site wave (fill: outcome). Grey zone below 25 degrees: approaching close-out (Lewis et al., 2015); red line: the 30 degrees the author associates with good surfers (rating-7 limit 29 degrees).

4.5 Reef size and volume

The author's observation (a) is that reefs need a large volume and that small reefs such as Pratte's do not work. Volume ranks with outcome (Figure 6): ρ = 0.72 (p = 0.021, n = 10) for the stated volume of the built reefs and 0.93 (p = 0.005, n = 8) for the model volume. All three reefs that do not work are 4,900 m³ or smaller and both built reefs that work are 25,000 m³ or larger, because volume buys relief and ramp length: a 1:16 to 1:20 ramp on a 4 to 5 m bed needs 15,000 to 25,000 m³ per 100 m (Section 7). Burkitts, a natural reef reshaped by moving about 300 m³ of rock, works, so volume matters when the reef must create the whole relief; and size is collinear with material and budget, so the association is descriptive. The direction is in the literature: "the single largest limitation to building a successful artificial reef in terms of surfing is the issue of scale" (Shand, 2011, as cited in Mead & Borrero, 2011, p. 2), and Pratte's failure was attributed to its "small size" (Borrero & Nelsen, 2003, pp. 13-14). Volume is not sufficient: Narrowneck, the largest reef, works only rarely (Mead & Borrero, 2011, p. 2).



Spearman rank correlation with outcome:
built reefs: ρ = 0.72, p_{perm} = 0.021 (n = 10)
incl. Burkitts*: ρ = 0.43, p_{perm} = 0.206 (n = 11)
modelled volume: ρ = 0.93, p_{perm} = 0.005 (n = 8)

* Burkitts: natural reef reshaped (about 300 m^3 of rock moved)

stated volume (source) volume of the 3D model

Figure 6. Reef volume against outcome for the eleven projects with a known outcome (filled: volume stated in the sources; open, joined by a line: volume of the 3D model). Burkitts (*) works at 300 m³ because it is a reshaped natural reef. Spearman ρ with the ordinal outcome (works 3, works rarely or mixed 2, does not work 1): 0.72 (permutation p = 0.021, n = 10) for the stated volume of the built reefs, 0.43 (p = 0.21, n = 11) with Burkitts, 0.93 (p = 0.005, n = 8) for the model volume; descriptive only. Sources: Appendix C.

4.6 Surfable window and tide

Reef waves work within narrow windows of wave height and water level. Pratte's broke "1-5 times per year" (Borrero & Nelsen, 2003, as cited in Rendle, 2016, p. 39), which local witnesses confirm anecdotally: it "never produced a surfable wave" in one account and was "really tide sensitive and wave height sensitive" in another (viewer comments, grade D; Dan Harmon, 2026, retrieved 19 Oct 2026); and Palm Beach works for easterly or southerly swells above H_s 0.8 m with peak periods above 7 s (Prenzler et al., 2023); the designers attribute the narrow windows to small volume (Mead & Borrero, 2011, pp. 2, 6). This supports the first part of observation (d), that reef waves are active on few days.

The computation steps the water level through the tidal range and counts a level as surfable when the wave breaks on the reef with $\xi_b < 2$ and $Y \geq 1.6$, weighted by a synthetic spring-neap tide (Appendix A). Given a suitable swell, Palm Beach, Cables and Albany are surfable 0.62 to 0.70 of the time, Borth 0.27, Mount Maunganui and Narrowneck 0.15 to 0.16, Pratte's 0.04, Bunbury 0.01 and Boscombe never (Table 2). Under the Israeli tide (spring range 0.40 m; Rosen, 2001) the average does not rise (mean change +0.01; 66 % of cases change by less than 0.05), but the outcome becomes a function of wave height alone: for a given swell the reef works all day or not at all (Figure 7). The Palm Beach geometry under a 1 m wave rises from 0.69 to 1.00, while Mount Maunganui, which breaks only at low water, falls from 0.41 to 0. A synthetic Haifa-type reef keeps 0.87 to 0.98 of its no-tide surfable time at a 0.2 to 0.4 m range, against 0.64 to 0.68 at 4.3 m. The tidal control is documented: before its renewal Narrowneck broke regularly above an RMS height of about 0.8 m at low tide and 1.2 m at high tide (Corbett et al., 2023, p. 668), and the Albany design relied on a "low tidal range" (City of Albany, 2023, p. 2). Without a wave climate, the fractions are shares of the time with a suitable swell, not days per year.

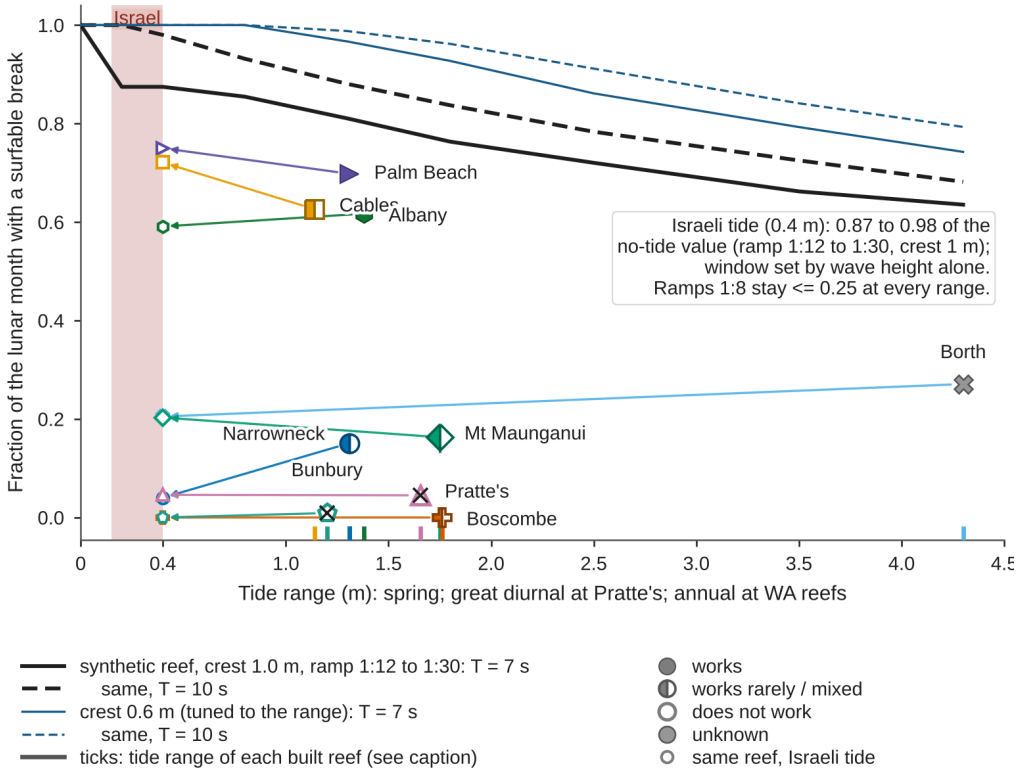


Figure 7. Fraction of the lunar month with a surfable break, given that the swell is present, against the tide range (local ranges as in Figure 1). Lines: a synthetic Haifa-type reef (1:50 bed, crest 10 m wide), mean over H_0 = 0.6 to 1.5 m, for a crest 1.0 m below MSL (black) and the best crest depth at each range (blue), at T = 7 s (solid) and 10 s (dashed). A break counts as surfable when it occurs on the ramp or crest with ξ_b < 2 and $Y \geq 1.6$ (Battjes, 1974; Mead & Black, 2001b), on a synthetic semidiurnal spring-neap tide. Symbols: the nine modelled reefs at their own tide range (fill: outcome), with an arrow to the same geometry under the Israeli tide (spring 0.40 m, neap 0.15 m; Rosen, 2001). No wave climate is included, so the fractions are not days per year.

4.7 Synthesis

The crest depth decides which waves break and the ramp gradient how they break; every reef refracts its waves to peel angles for advanced surfers; and a small tide makes the surfable time depend on wave height alone (Table 2). The model is one-dimensional with linear shoaling, ignores refraction and focusing across the reef (except for the peel angle) and rests on idealised geometry for six of the nine reefs, so its classes are indicators, not predictions of ride quality.

Table 2. Computed wave indicators per reef (derived in this review; Appendix A). Crest line at MSL unless noted. H_0 window: smallest H_0 that breaks on the reef and largest that does not break seaward of it (site period, MSL). ξ_b and Y: medians over all cases that break on the reef (H_0 0.5-3 m, T 6-14 s, three tide levels), in brackets the share with Y < 1.6. α_b: refracted peel angle for H_0 0.5-3 m and the Hutt rating range. Surfable: fraction of time with a suitable swell under the local and the Israeli tide (at Cables, Bunbury and Albany the local tide is the 2026 predicted annual range, a wider band than the spring range). Volume: stated / model (m³).

Reef (outcome)	Crest depth (m)	Depth gap (m) / ramp (m), slope	H_0 window (m)	ξ_b	Y (Y < 1.6)	α_b (deg) / rating	Surfable, local / Israeli	Volume (m³)
Palm Beach (works)	1.50	5.49 / 89, 1:17	0.81-5.16	0.49	2.01 (0.00)	16-35 / 7-9	0.70 / 0.75	25,000 / 21,948
Albany (works)	1.17	6.15 / 22, 1:4.0 (centreline 1:23.7)	0.52-4.77	0.75	1.64 (0.47)	22-51 / 5-9	0.62 / 0.59	34,000 / 33,999
Cables (mixed)	1.56	4.87 / 44, 1:20	0.78-3.44	0.63	1.69 (0.39)	23-37 / 7-9	0.63 / 0.72	5,500 / 4,683
Narrowneck (rarely)	1.92	2.08 / 16, 1:15 (second line 3.92 / 37)	1.10-4.15	0.97	1.36 (0.72)	25-57 / 4-9	0.15 / 0.04	70,000 / 13,496
Mt Maunganui (rarely)	2.08	2.83 / 20, 1:9.3	1.16-3.04	1.62	1.14 (0.91)	21-52 / 5-9	0.16 / 0.20	2,800 / 2,943
Boscombe (rarely)	0.90	4.51 / 13, 1:3.1	0.48-4.09	2.00	1.06 (1.00)	25-57 / 4-9	0 / 0	12,800 / 13,112
Pratte's (does not work)	1.86	1.60 / 5, 1:4.1 (centreline)	1.31-2.28	1.13	1.37 (0.63)	22-48 / 6-9	0.04 / 0.05	1,350 / 442
Bunbury (does not work)	0.68	2.31 / 5, 1:2.7	0.28-1.62	2.97	1.05 (1.00)	dome, not computed	0.01 / 0	105 / 116
Borth (unknown)	crest 1.09 above MSL	5.35 / 73, 1:14.5	0.05-3.03	1.06	1.39 (0.77)	26-62 / 3-9	0.27 / 0.20	30,119 / 19,954

Notes: Pratte's crest line gives a 1 m ramp (a tip artefact); its centreline is shown. Narrowneck's stated volume is "some 70,000 m³" (Mead & Borrero, 2011, p. 2); the model covers the visible container field only. Borth values come from the analysis model (crest line over the highest rock, 1.09 m above MSL; model volume of the 7,435 m² analysis footprint), whereas Table 1 gives the reef crest (0.19 m above MSL) and the volume of the 2022 laser-survey edge (30,119 m³, 8,675 m²); the outlines differ (Appendix C, note 3). Y on slopes steeper than 1:9 is extrapolated from the Mead and Black (2001b) fit (share of cases outside the fit range: 0 at Palm Beach, 0.34 at Cables, 0.57 to 1.00 at the other reefs).

5 Sediment transport and coastal stabilisation

Most surf reefs were promoted as multi-purpose structures that would also stabilise the beach. This section compares the standard shoreline predictors with the surveyed response and asks how a surf reef can be combined with coastal stabilisation.

5.1 Theory: transmission, salients and erosion in the lee

A submerged reef dissipates part of the incident energy by breaking and transmits the rest. For low-crested structures d'Angremond et al. (1996) fitted

$$K_t = -0.4\frac{R_c}{H_i} + c\left(\frac{B}{H_i}\right)^{-0.31}\left(1 - e^{-0.5\xi}\right), \qquad 0.075 \le K_t \le 0.8,$$

with freeboard R_c (negative when submerged), crest width B, surf similarity ξ and c = 0.64 (permeable) or 0.80 (impermeable); Briganti et al. (2003) give a variant for wide crests (both as reproduced by Blacka et al., 2013, Eqs. 25-29). The transmitted energy is K_t^2.

The shoreline responds to gradients of the wave-driven flow in the lee. In the models of Ranasinghe et al. (2006), a reef close to shore drives diverging currents and erosion, and a reef farther offshore drives two converging cells and accretion (as summarised by Rendle, 2016, p. 43); erosion is likely when the reef's distance from the shoreline is less than the natural surf-zone width ($S_a/SZW < 1$) and accretion when $S_a/SZW > 1.5$, peaking near 2 (as reproduced by Blacka et al., 2013, pp. 22-23). Reviewing built structures, Ranasinghe and Turner (2006, abstract) found that "contrary to expectations, a majority of the submerged structures constructed to date have resulted in shoreline erosion in their lee". The empirical method of Black and Andrews (2001) predicts little change for an alongshore length to offshore distance ratio B/S below about 0.1 and a salient amplitude from $X_{off}/B = 0.5\,(B/S)^{-1.27}$ (Eq. 30, as reproduced by Blacka et al., 2013, p. 21); Blacka et al. (2013, p. 22), citing an earlier evaluation by Evans and Ranasinghe, note that the method over-predicts at built reefs.

5.2 Predictions for the built reefs

Table 3 applies these predictors to the reconstructed reefs. At mid tide the submerged reefs transmit 41 to 80 % of the site-wave height, i.e. 17 to 64 % of its energy; only Borth, whose crest dries, blocks most waves. Both predictors expect a salient at every reef (B/S 0.23 to 0.78; amplitude 11 to 114 m; S_a/SZW 1.5 to 5.5 under typical waves), but storms widen the surf zone until $S_a/SZW = 1$ at H_0 of 1.5 to 4.5 m, putting every reef in the erosion-likely regime.

Table 3. Predicted and observed shoreline response (predictions derived in this review; observations from Section 5.3). K_t for the site wave at low, mid and high tide (d'Angremond et al., 1996; Briganti et al., 2003); B/S and the salient amplitude of Black and Andrews (2001, Eq. 30), in brackets the lower reading of their Eq. 31 (with an assumed total salient length of 1 to 3 B); S_a/SZW for the site wave and twice the site wave (Ranasinghe et al., 2006).

Reef	K_t low / mid / high	B/S	Predicted salient (m)	S_a/SZW (H / $2H$)	Observed lee response
Boscombe	0.05 / 0.48 / 0.47	0.39	84 (Eq. 31: 11-35)	5.5 / 3.7	+3.4 m (95 % CI +1.3 to +5.3), before/after with controls
Palm Beach	0.29 / 0.52 / 0.75	0.43	114	4.8 / 2.0	2 m or less (chart, low confidence)
Cables	0.32 / 0.46 / 0.60	0.44	102 (Eq. 31: 14-41)	3.8 / 2.1	lee high pre-dates the reef; beach bulge shrinking
Borth	0.05 / 0.05 / 0.43	0.38	89	2.2 / 1.4	+30 m bulge (+12 to +48), single survey, no baseline
Pratte's	0.80 (upper limit)	0.63	40	2.8 / 1.3	+13 m in 5 months, no control; "no discernible impact"
Narrowneck	0.42 / 0.64 / 0.68	0.39	104	3.3 / 1.3	not testable (continuous nourishment)
Mt Maunganui	0.44 / 0.66 / 0.80	0.23	79	2.1 / 1.1	"subtle and within the natural variation"
Albany	0.20 / 0.41 / 0.63	0.78	77	2.4 / 1.6	no post-construction data (opened 2025)

Reef	K_t low / mid / high	B/S	Predicted salient (m)	S_a/SZW (H / 2H)	Observed lee response
Bunbury	0.46 / 0.64 / 0.80	0.34	11	1.5 / 0.7	removed within days

5.3 Observed response: the seabed and beach surveys

Only five of the 13 reefs have measured seabed or beach data that can test a reef effect (Appendix A). For each, this review compared the lee with control stretches 1.5 to 3.5 reef lengths alongshore and, where possible, before with after.

Boscombe has the best-controlled record, 316 beach-profile surveys from 2004 to 2026 (Channel Coastal Observatory, 2026a). Relative to four control lines, the 0 m ODN contour behind the reef advanced by +3.4 m (95 % confidence interval +1.3 to +5.3 m) after construction (Figure 8): 0.04 of the 84 m predicted by Eq. 30, while beach management moved the whole frontage by +72 m³/m in 2006-07 and +40 m³/m in 2020-21. This agrees with Rendle (2016, p. 102): “no significant shoreline response”. The designers’ 60 m salient after six months (Mead & Borrero, 2011, p. 5) coincided with 11,100 m³ of surplus fill sand left on the beach and dispersed after the first winter (Rendle, 2016, pp. 125-128).

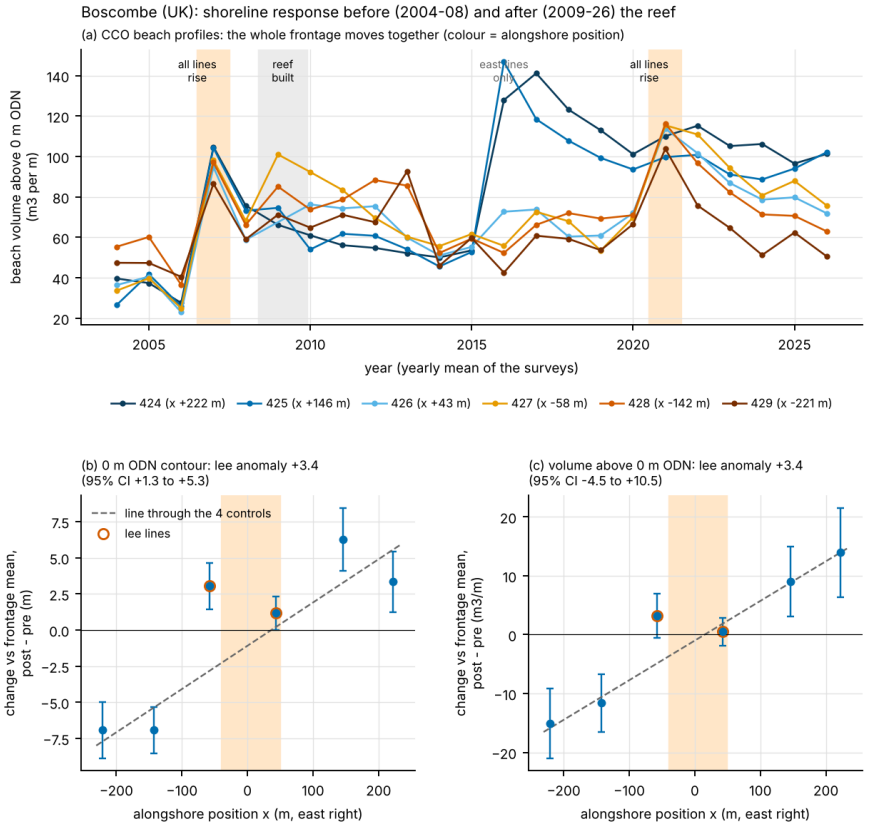


Figure 8. Shoreline response at Boscombe before (2004-2008) and after (2009-2026) the reef, from the beach-profile surveys of the Channel Coastal Observatory (2026a), analysed in this review. (a) Beach volume above 0 m ODN on six profile lines (yearly means; colour: alongshore position x): the whole frontage rises together in 2006-07 and 2020-21, a beach-management signal. (b) Change of the 0 m ODN contour (post minus pre, relative to the frontage mean) against alongshore position: the two lee lines (rings) lie +3.4 m (95 % confidence interval +1.3 to +5.3 m, bootstrap over survey dates) above the line through the four control lines. (c) The same for the beach volume: +3.4 m³/m (−4.5 to +10.5). Black and Andrews (2001, Eq. 30) predict a salient of 84 m.

Cables, the only reef with offshore soundings before and after construction (Western Australia Department of Transport, 2020), has a lee seabed 0.60 m higher than the controls, but the offset already existed in 1995 (Figure 9): the lee bed changed by +0.03 m (standard error 0.04 m) from 1995 to 2002, and the beach bulge shrank from +26 m (2001) to +10 m (2012), whereas a reef-driven salient would grow.

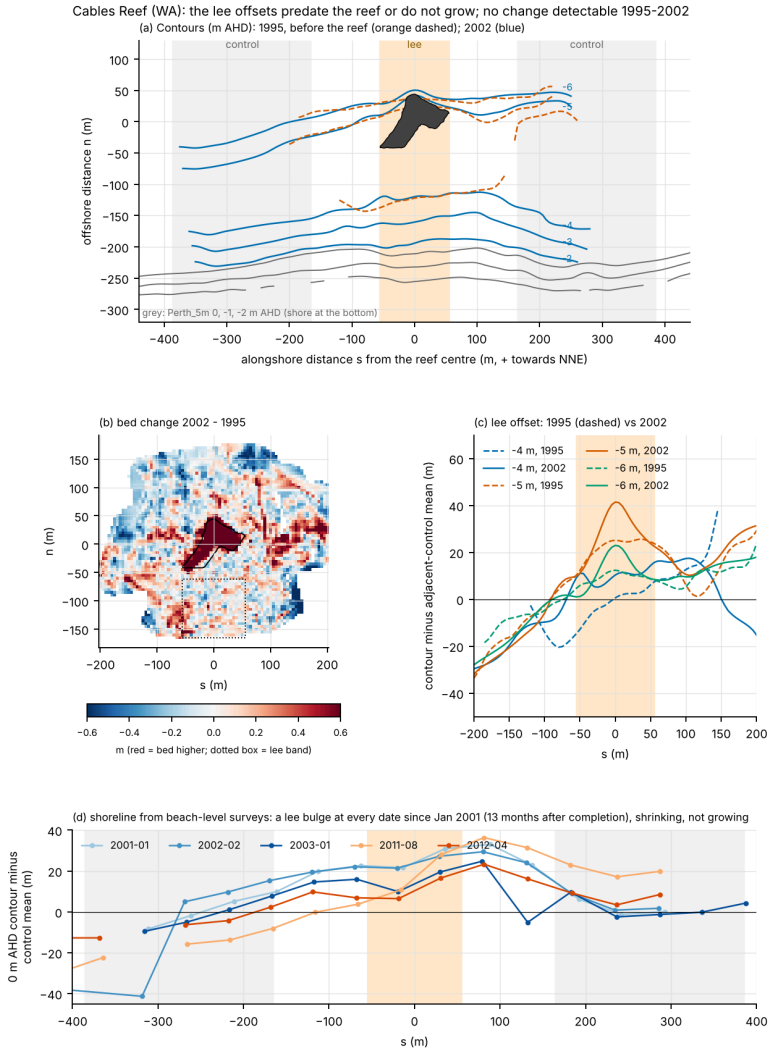


Figure 9. Seabed and shoreline around Cables Reef from the single-beam soundings of the Western Australia Department of Transport (2020) and beach-level surveys, analysed in this review. (a) Contours (m AHD) in December 1995, before the reef (dashed), and February 2002 (solid), with the reef footprint, the lee band (orange) and the control windows (grey). (b) Bed change 2002 minus 1995 (red: bed higher). (c) Offset of the −4, −5 and −6 m contours in the lee relative to the controls in 1995 (dashed) and 2002 (solid): the offset pre-dates the reef. (d) Shoreline (0 m AHD) relative to the control mean, 2001-2012: a lee bulge that shrinks from +26 m to +10 m rather than growing.

Borth gives the one positive signal: in the 2022 lidar survey (Welsh Government, 2022) the 0 m contour bulges +30 m behind the reef (+12 to +48 m depending on the control; Figure 10), but without a pre-construction survey, and within a scheme of groynes, breakwaters and shingle recharge, the bulge cannot be attributed to the reef. At Palm Beach chart contours are offset by 2 m or less (low confidence).

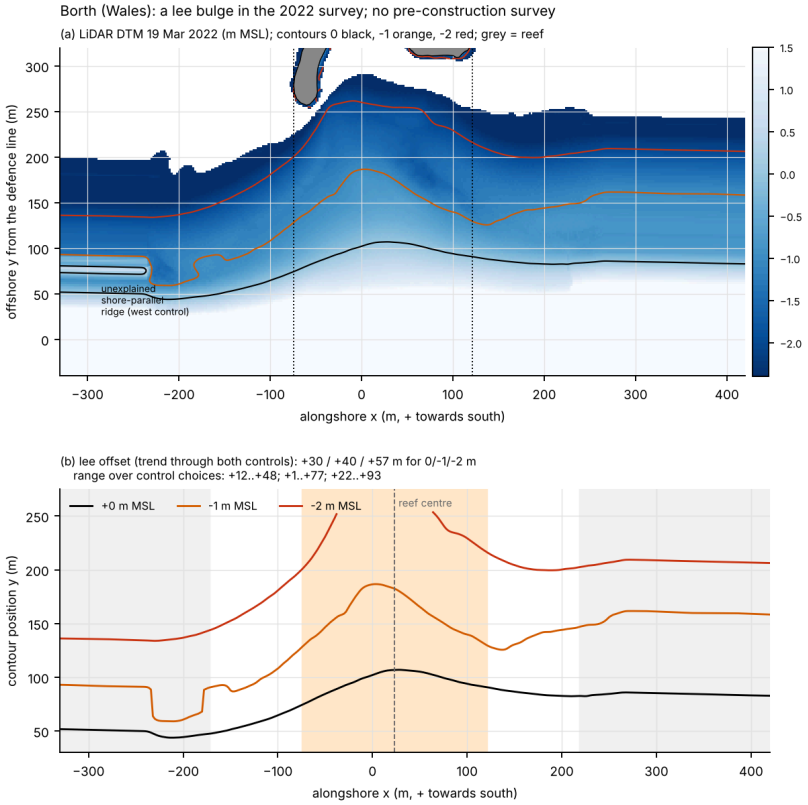


Figure 10. Seabed behind the Borth reef from the lidar terrain model of 19 March 2022 (Welsh Government, 2022), analysed in this review. (a) Bed level (m MSL) with the 0, −1 and −2 m contours and the two reef mounds (grey). (b) Contour positions alongshore, with the lee window (orange) and the control windows (grey): relative to the trend through both controls, the lee contours lie +30, +40 and +57 m farther offshore (+12 to +48, +1 to +77 and +22 to +93 m over control choices). A single survey without a baseline cannot separate the reef from the groynes, breakwaters and shingle recharge of the same scheme.

The literature agrees. At Mount Maunganui effects were “subtle and within the natural variation” with “no noticeable salient” (Weppe et al., 2007, as cited in Blacka et al., 2013, p. 103); at Pratte’s monitoring found “no discernible impact on either the surrounding bathymetry or the beach” (Blacka et al., 2013, p. 117); and about half of the protection structures reviewed by Blacka et al. (2013, p. 73) had no significant accretionary impact. The one positive report, a wider beach at Kovalam, rests on the designers’ photographs over less than three years (Mead & Borrero, 2011, p. 5; Blacka et al., 2013, p. 70). The observed lee response is thus 0.02 to 0.06 of the Black and Andrews prediction, because transmission is high, the reefs are short (12 to 132 m alongshore) and diffract the waves, and beach management produces signals an order of magnitude larger.

5.4 The Gold Coast and the nourishment confound

The author’s observation (b) is that the Gold Coast reefs are claimed to protect the coast although the shoreline shows little change, and that the claim cannot be attributed to the reef because sand is nourished continuously. The record supports it. Narrowneck was built as “a coastal control point to stabilize the nourished beaches to the south” after 1.3 million m³ of nourishment in 1999-2000, with at least 60,000 m³ a year planned thereafter (Jackson et al., 2012, pp. 1-2). The designers report a beach 30 to 50 m wider (Mead & Borrero, 2011, p. 4), whereas Blacka et al. (2013, pp. 98-100) found “only a minor planform perturbation” of about 10 m and called the published widening “a likely misreading” of an arbitrary reference line, and Jackson et al. (2012) found “a minor groyne effect” and no long-term salient. Palm Beach was part of an AU$18 million nourishment project that placed about 470,000 m³ on the beach (Royal HaskoningDHV, n.d.); the City’s statement that the reef “retained about 550,000 cubic metres” in two years (ABC News, 2022) is of the same order as the volume placed, and there is no nourishment-only control (Prenzler et al., 2023).

5.5 Burial, settlement and scour

What reefs change reliably is the bed near the structure: “vertical displacement of the structures of the order of 0.5 to 2 m as well as localised bed scouring” at most of them (Blacka et al., 2013, p. 72). At Pratte’s the bed scoured to about 4 m, the crest dropped 1.5 m in a year and many bags were buried at removal (Blacka et al., 2013, pp. 116-117); at Narrowneck a storm bar buried the seaward containers (Jackson et al., 2012, pp. 4-5; Corbett et al., 2023, p. 663); at Boscombe the −3.5 m contour moved 50 m shoreward in the lee (Rendle, 2016, p. 102). The rock reef at Cables, on limestone bedrock, shows no toe scour or burial (this review). Settlement and burial remove relief, which Section 4 found decisive for surfing; Section 7.4 quantifies the effect for Haifa.

5.6 Combining a surf reef with coastal stabilisation

The two functions pull the design apart. A surfing crest lies about 1 to 2 m below MSL (Table 2) and therefore transmits 17 to 64 % of the energy, and a short reef changes the planform little; placing it at about twice the surf-zone width favours accretion under typical waves (Ranasinghe et al., 2006), but not in storms. The Gold Coast schemes therefore combine nourishment with a reef acting as a control point (Jackson et al., 2012; Prenzler et al., 2023); the engineers' claim that over 30 years the reef "definitely stacks up against beach nourishment" (ABC News, 2022) has not been tested against a nourishment-only baseline. A surf reef can hold nourished sand and add amenity within a stabilisation scheme, but it should not be relied on as stand-alone protection, and a protective effect should be claimed only from before/after surveys with controls and a nourishment record, which no reef examined has published in full.

6 Materials and durability

Surf reefs have been built from geotextile sand containers (GSCs, also called geotubes or bags), a sand-filled rubber bladder (the Airwave), quarried rock and concrete. This section compares their record, the wave loads in the breaking zone, and their ecology and cost.

6.1 Failure record

The author's insight (c) holds that soft structures do not survive long in the breaking zone. Table 4 summarises the fate of the reefs by material: six of the seven geotextile or inflatable reefs suffered major damage or loss within a few years, and the seventh, Narrowneck, survives only through continuous maintenance; none of the four rock reefs failed (Fisher exact test, $p = 0.015$; derived in this review). The comparison is confounded, because the rock reefs are newer and better funded. The designer of several of these reefs now calls the sandbags "an evolutionary deadend" (Reid and Sumner, 2021) and writes that "the fatal flaw of the bags was their propensity for failure" (Mead, as quoted in Geosynthetics Magazine, 2023).

Table 4. Material and fate of the surf reefs (sources as cited).

Reef (year)	Material	Fate	Named cause
Narrowneck (2000)	about 400 GSCs, 20 m long, 3-4.5 m diameter (Blacka et al., 2013, p. 97)	containers compromised and seaward units buried by 2011; 84 containers added 2017-18 (Jackson et al., 2012; Corbett et al., 2023)	anchors, propellers, spears and knives (Jackson et al., 2012, p. 7)
Pratte's (2000)	woven polypropylene and polyester bags, 3.0 x 2.1 x 1.2 m	crest down 1.5 m in a year; fabric "could be torn by hand"; removed 2008-2010 (Blacka et al., 2013, pp. 116-117)	settlement, scour, degradation of the fabric
Mount Maunganui (2008)	large GSCs, up to 7.2 x 3.2 m in section	2,800 of 6,500 m³ in place (Mead and Borrero, 2011, p. 3); largest containers removed 2014 (Rendle, 2016, p. 41)	ruptured containers and unsealed ports during construction
Opunake (2008, incomplete)	27 GSCs	never completed (Taranaki Regional Council, 2009)	funding and construction
Boscombe (2009)	about 55 GSCs, the largest 65.5 m long	fill reworked ("pillowing") in the first winter; one container lost in 2011; closed 2011 (Rendle, 2016, pp. 94-98)	contested: weakness, propeller strike or vandalism
Kovalam (2010)	GSCs, about 4,000 m³	bags washed ashore after the first monsoon (Rendle, 2016, p. 42)	debris strikes (designer's view)
Bunbury Airwave (2019)	Hypalon bladder filled with sand (and air)	seam tore during installation; removed within days (ABC News, 2019)	seam glued because the Hypalon could not be welded (Bunbury Mail, 2022)
Cables (1999)	rock, 1.5 t and 3 t, on limestone bedrock	in place; crest change −0.16 m in 2.2 years, within survey error (Appendix C)	none reported
Borth (2012)	graded rock, units of several tonnes	in place	none reported
Palm Beach (2019)	rock, about 25,000 m³	in place (Prenzler et al., 2023)	none reported
Albany (2025)	rock in three graded layers	core rock moved in a storm before armouring (Appendix C); 18 rocks repositioned for safety (Sheil, 2026)	storm during construction; surfer injuries

The GSC failures follow a common sequence: point damage from anchors, propellers, knives, debris or construction faults, then loss of fill and change of shape, because "as soon as even a single sand-filled container broke and lost sand, the whole structure changed shape due to the fluid nature of sand" (Mead, as quoted in Geosynthetics Magazine, 2023). At Boscombe the damage to one large container led to the loss of two, 12.5 % of the reef's volume, and the reef was judged "unlikely" to survive its proposed 25-year life "without multiple repairs" (Rendle, 2016, pp. 98, 121). The manufacturers' design life of about 20 years for outer-layer containers (Geofabrics, 2013, as cited in Blacka et al., 2013, p. 35) contrasts with one to eight years before major loss in the field.

The Airwave tore along a glued seam near completion in December 2019 (ABC News, 2019); the Hypalon rubber could not be welded (Bunbury Mail, 2022). Its founder states that the final design was filled with sand instead of air, which did not respond well to the waves, that the reef made perfect waves until a seam opened a couple of days after installation, and that the next version will be concrete

(T. Bottegal, Airwave founder, pers. comm., 8 Oct 2026). The timing differs from the press record and the wave quality is unverified; both accounts agree that the seam, not the rubber, failed.

6.2 Wave loading in the breaking zone

Bar and Drimer (2023) computed the loads on a submerged, impermeable geotube with a numerical wave flume and defined a sliding index $SI = F_y/(W' - F_z)$ (horizontal force F_y, uplift F_z, submerged weight W') and a variant SI_c that adds the uplift of the dynamic pressure on the contact area. They found that "the most critical state, regarding the sliding stability of the Geotube, occurs, typically just before breaking" (p. 1), when an up-wave horizontal load coincides with uplift (p. 8), and advised assuming that the dynamic pressure penetrates under the tube (p. 10). They name "too-small size Geotubes" and "sand coverage" as the common causes of failure (p. 2). The tube is a rigid body in the model, which gives forces but not fabric stress or tearing. A later paper by the same author on tube tearing, mentioned to this review, could not be located; the analysis rests on Bar and Drimer (2023).

6.3 Stability calculations

This review estimated the loads by integrating the linear incident-wave pressure over a rigid tube section, calibrated against values read from Bar and Drimer (2023), and bounded the uplift ratio $U = \max F_z/W'$ between a sealed base (upper bound) and a base reached by the wave pressure (U_c, lower bound) (Appendix A). Under depth-limited breakers ($H_b = 0.78\,h$), tubes 1.0 to 1.65 m high reach $U = 1$ at base depths of 2.3 to 3.9 m and tubes 2.2 to 3.2 m high at 5.3 to 8.0 m (Figure 11a): the sealed-base uplift scales as $U \approx H/(2c\,\Delta D \cosh kh)$ (D tube height, Δ relative submerged density, $c \approx 0.87$), so small tubes lift first. If the pressure reaches the base, $U_c \le 0.22$ and the calibrated SI_c of 0.37 to 0.56 makes sliding marginal against a friction coefficient of about 0.5 (0.48 measured bag on bag by Corvaro et al., 2022). The single-tube criterion of van Steeg et al. (2011), $H_s/(\Delta\sqrt{BD}\,(f\cos a + \sin a)) \le 0.65$, gives critical heights of 0.34 to 1.65 m for the built reefs' tubes, which depth-limited seas ($H_s \approx 0.55\,h$) reach at base depths of 0.6 to 3.0 m, inside every surf-reef crest. By the Hudson formula with $K_D = 2$ (US Army Coastal Engineering Research Center, 1984, as tabulated by Verhagen, 2012), rock of 1 to 2 t resists the same waves; the Albany armour (D_{50} 0.94 m) is stable to H_s 2.1 to 2.5 m but its core rock only to 1.1 to 1.3 m, consistent with the storm movement of exposed core rock during construction. A tube at the break sees 7,200 to 14,400 load cycles a day.

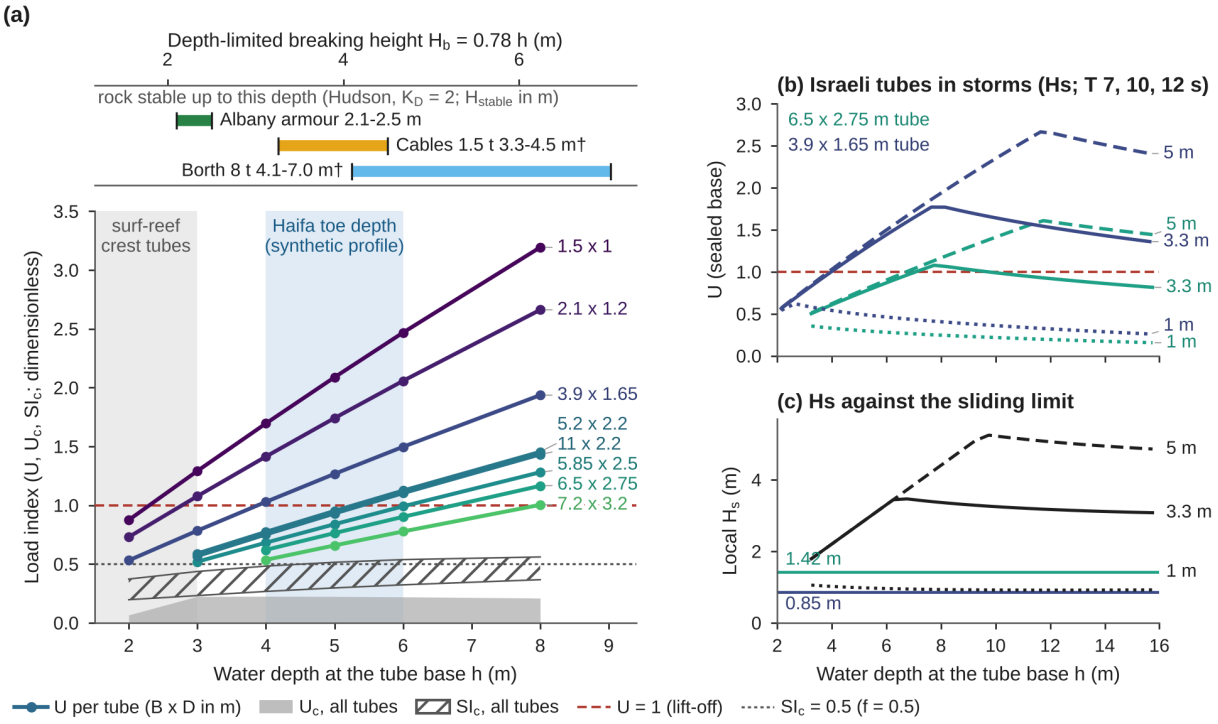


Figure 11. Estimated load on sand-filled geotextile tubes and rock in the breaking zone (an estimate; tearing is not modelled). (a) Sealed-base uplift U (maximum vertical force over the wave cycle divided by the submerged weight; an upper bound) per tube size $B \times D$ under depth-limited breakers ($H_b = 0.78\,h$, T = 6 to 12 s), with the range of U_c if the pressure reaches the base (grey) and of the calibrated sliding index SI_c (hatched); bars above give the depth to which the rock of Albany, Cables and Borth stays stable (Hudson formula, $K_D = 2$). (b) U for the Israeli 6.5 x 2.75 m and 3.9 x 1.65 m tubes in storms of H_s = 1, 3.3 and 5 m, and (c) the local H_s (limited to $0.55\,h$) against the single-tube sliding limit of van Steeg et al. (2011). Forces from a linear Froude-Krylov model of a rigid section calibrated against values read from Bar and Drimer (2023), not their numerical model; details in Appendix A.

For the author's insight (e), that most wave power acts at the break and lifts the tube up and then down, which is why most tubes tear, the load part is supported: the critical state occurs just before breaking with simultaneous uplift (Bar and Drimer, 2023), and small or sealed tubes are lifted by ordinary breakers. The tearing part is not modelled by any source found, and "most tubes tear" is not supported, since the reported causes are point damage, undersizing, burial and construction faults (Table 4). Repeated lifting and

flapping may fatigue fabric and seams (“the initial cause of failure is masked by the damage suffered during cyclic flapping and deflation of the container”; Jackson et al., 2012, p. 7), but this is a hypothesis that needs fabric and seam tests. Insight (c), that soft structures do not survive long in the breaking zone, is supported by both the record and the stability numbers.

6.4 Ecology and cost

Insight (c) also holds that geotubes support marine life and are the cheapest structure. Colonisation of geotextile is documented (seaweed on Korean geotubes after one year; Oh and Shin, 2006a), and the designer chose containers partly as a “good substrate for marine organisms” (Geosynthetics Magazine, 2023), whereas at Ashkelon a surfers’ group stated that “nothing develops near the geotubes” (Rinat, 2019). Rock and concrete are colonised too, and bio-enhanced concrete is reported by its maker to more than double invertebrate species richness on a Haifa port breakwater (ECOncrete, n.d.; company data). No comparative study was found, so the ecological advantage of geotubes is a new, untested observation.

On capital cost the containers are cheaper: A$56 per m³ at Narrowneck against A$457 for the Cables rock (Blacka et al., 2013, Table 4.4, 2013 AUD), although the small or incomplete GSC reefs cost A$274 to A$545 per m³. Over the life of the structure the advantage shrinks: renewal (Narrowneck), removal (Pratte’s, about $551,000, slightly more than its construction; Blacka et al., 2013, p. 119), plastic debris (Rendle, 2016, p. 122) and a short field life. “Cheapest” is supported for construction cost only; for a reef in the breaking zone, durability decides the choice (Section 7).

7 Implications for the Israeli coast and Haifa

This section applies Sections 4 to 6 to the Israeli Mediterranean coast and to Haifa, where a surf reef is being considered.

7.1 Tide, water level and waves

The semidiurnal tide has a range of about 0.4 m at springs and 0.15 m at neaps along the whole coast (Rosen, 2001; Spanier and Zviely, 2022), smaller than at any built reef (1.1 to 4.3 m; Section 1.3). Weather and season move the water level more: the 1-year low and high levels are −0.38 and +0.64 m, and the 100-year high is +1.10 m (Rosen, 2001, Table 1). The most common H_s is 0.5 to 1.0 m; summer is calm, winter brings about five storms a year above H_s 3.5 m at Haifa, and 70 % of waves come from west to west-north-west (Spanier and Zviely, 2022; Bitan and Zviely, 2020); storm waves above 5 m have periods up to 15 s (Carmel et al., 1985). Haifa Bay is sheltered: the nearshore wave exceeded 12 hours a year is 3.0 to 3.3 m against 5.1 to 5.8 m on the open coast, and the closure depth 4.7 to 5.2 m (Bitan and Zviely, 2020). The author’s Shikmona buoy readings (summer 2025, a three-week sample) put the smallest surfable sea at H_s 0.45 to 0.60 m with peak periods of 5.7 to 6.2 s from 284 to 294 degrees (author’s observation). Israeli surf is thus below the Palm Beach window (H_s above 0.8 m, T_p above 7 s; Prenzler et al., 2023) and close to Boscombe’s design range (H_{10} = 1 ± 0.5 m, T = 7 ± 2 s; Mead et al., 2010), where a shallow crest built for small waves gave a collapsing wave (Section 4.3); the short periods help, because they soften the breaker on a given ramp.

7.2 A small tide: consistency, not more surf

Insight (h-A) holds that the small tide would make a reef wave more consistent and narrow the range of loads. The first part is supported (Section 4.6, Figure 7): a synthetic Haifa-type reef keeps 0.87 to 0.98 of its no-tide surfable time, and the break depends on wave height alone, although the average surfable time of the built geometries does not rise. The second part is only partly supported. The tidal range of the depth-limited crest wave is 0.31 m in Israel against 0.9 to 3.4 m at the built reefs, but damaging loads come from storms: the storm and seasonal water-level range (1.02 m) is 2.5 times the spring tide, Haifa Bay storm seas of H_s 3.0 to 3.3 m exceed the van Steeg et al. (2011) limit for single tubes by a factor of 2 to 5, and a sealed-base 6.5 x 2.75 m tube is lifted at 7 to 9 m depth (Figure 11). The material, not the tide, decides durability.

7.3 Israeli experience with geotextile tubes

The geotubes that tore in Israel were at Ashkelon North; no geotube was built at Netanya, which is protected by rock breakwaters (State Comptroller of Israel, 2020, p. 71; 2024, p. 48). At Ashkelon North 48 submerged geotubes were laid in five clusters between July 2017 and September 2019 for about NIS 11 million, as temporary cliff protection (State Comptroller of Israel, 2020, pp. 71, 83). Several “were damaged and torn shortly after being laid in the sea, after the winter storms, and had to be repaired” (State Comptroller of Israel, 2020, p. 71; translated), and in early 2019 severe damage of unknown cause halted the work (Rinat, 2019). Twelve tubes were replaced in 2021, maintenance meant “sewing tears and collecting loose sheets”, and the protection of one section later collapsed (State Comptroller of Israel, 2024, pp. 29, 48, 53; translated). The official causes are “unprofessional laying, non-optimal filling, external damage for example from propeller blades, anchors or divers’ knives” (State Comptroller of Israel, 2024, p. 9; translated), and the agency estimates the life of the unarmoured sleeve used at months to a few years (p. 57). A senior Israeli coastal engineer argues that these failures do not prove geotubes unsuitable, since they can be strengthened and Haifa Bay waves are lower than those of the open coast (N. Drimer, as quoted in Zalul, 2025, p. 23; an expert opinion). The Israeli record agrees with the international one: the fabric is the weak point.

7.4 Sand mobility, burial and the sand-trap idea

The net drift runs northward, from about 400,000 m³ a year at Ziqim to 80,000 ± 20,000 m³ a year at the entrance to Haifa Bay (Spanier and Zviely, 2022); the Haifa Port breakwater trapped an average of 66,000 m³ a year from 1929 to 2004 (Zviely et al., 2009, p. 95). Gross transport is far larger: by the CERC formula (US Army Corps of Engineers, 2002), 1 m breakers at 10 degrees move about 2,700 m³ a day, the net annual drift of the southern Carmel coast (60,000 to 72,000 m³; Zviely et al., 2007) in about 22 days (derived; order of magnitude). A reef crest and toe within the 5 m closure depth lie in the active zone. The author has observed a seasonal bed-level change of about 1 m near Haifa; this is not yet published, but the seasonal shoreline swing of up to ±20 m (Zviely et al., 2009) on slopes of 1:70 to 1:30 implies 0.3 to 1.3 m (derived). A 1 m rise removes 19 to 40 % of the relief of a reef with a 0.8 to 1.5 m crest and a 4 to 6 m toe, and 1 m of settlement raises the wave needed to break on a 0.8 m crest from H_b 0.62 to 1.40 m, above the smallest surfable sea (derived). With geotextile reefs settling by 1 to 2 m (Section 5.5), burial is a central risk, as insight (h-B) anticipates.

The author proposes to trap sand in the high-sand seasons (October and May, by the author's observation) and to build the reef on it under a geotextile sheet. As a sand source the idea is not supported: at the observed salient ratio a reef holds about 10 to 2,400 m³, less than 4 % of one year's net drift, and even the Black and Andrews upper bound (21,000 to 102,000 m³) is a one-time fill, not a supply (derived). As a construction saving it is plausible: building on a bed 1 m higher under a 70 x 100 m footprint saves about 7,000 m³, half the fill of a 1:12 reef section (derived). No precedent was found and the sheet would face the uplift of Section 6.3, so the idea remains a proposal that needs a pilot.

7.5 Design envelope for Haifa and next steps

Without Haifa bathymetry, the envelope was screened on a synthetic profile (1:50 bed, crest 10 m wide, 200 m offshore) with the methods of Section 4, the Israeli tide and periods of 6 to 8 s (Figure 12; Table 5). Crests 1.0 m or less below MSL let at least 90 % of 0.6 to 1.5 m waves break on the reef; 1:8 ramps collapse the wave, 1:12 to 1:16 give an expert wave and 1:20 a mix of advanced and spilling waves; 1 m of settlement moves a 0.8 m crest to 1.8 m and cuts the surfable fraction from 1.00 to about 0.3. The largest uncertainty is the breaking criterion: the author's field observations (H_s 0.45 to 0.60 m breaking at 1.3 to 1.5 m depth, H_s/d about 0.35 to 0.40) imply earlier breaking than γ = 0.78 and would allow a crest at 1.2 to 1.5 m.

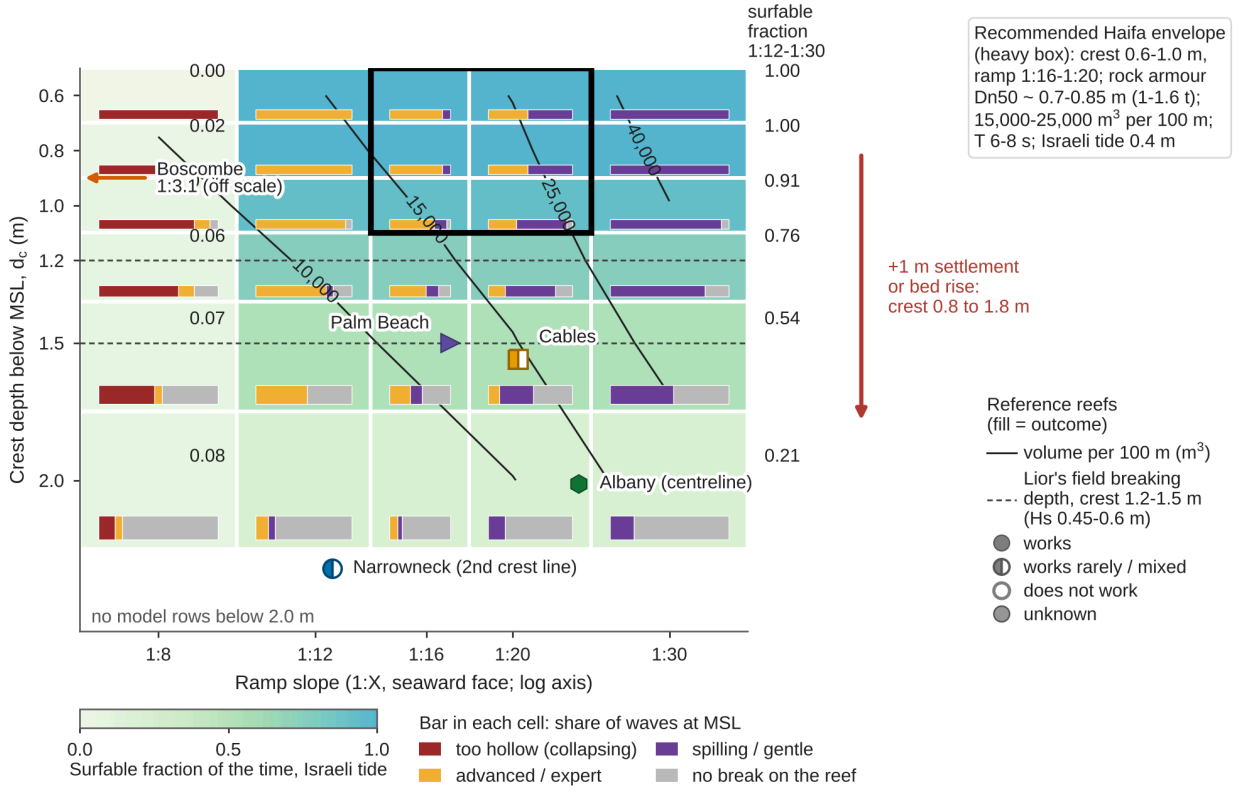


Figure 12. Design envelope for a surf reef at Haifa on a synthetic 1:50 bed (no Haifa bathymetry is available), by crest depth below MSL and seaward ramp slope (crest 10 m wide). Cell colour: fraction of the time the break is surfable under the Israeli tide (spring range 0.40 m); bars: share of waves (H_0 = 0.6 to 1.5 m, T = 6 to 8 s) that are too hollow, advanced or expert, spilling, or do not break on the reef (classes from ξ_b and Y; Battjes, 1974; Mead & Black, 2001b); contours: reef volume per 100 m. Heavy box: recommended envelope (crest 0.6 to 1.0 m, ramp 1:16 to 1:20, 15,000 to 25,000 m³ per 100 m); arrow: 1 m of settlement or seasonal bed rise; dashed lines: crest depths implied by the author's field breaking depths; symbols: built reefs (Boscombe, 1:3.1, off the scale). One-dimensional linear shoaling with γ = 0.78 and no refraction: a first screening only.

Table 5. Screening envelope for a Haifa surf reef (derived in this review; Sections 4 to 7).

Element	Recommendation	Basis
Crest depth	0.6 to 1.0 m below MSL (1.2 to 1.5 m if the field breaking depths hold)	at least 90 % of 0.6 to 1.5 m waves break on the reef
Seaward ramp	1:16 to 1:20; 1:30 for a beginner wave; never steeper than 1:12	breaker type and intensity (Section 4.3)
Size	100 m or longer; 15,000 to 25,000 m³ per 100 m (35,000 to 49,000 m³ at 1:30)	volume and outcome of built reefs (Section 4.5)
Material	rock armour, D_{n50} about 0.7 to 0.85 m (1.0 to 1.6 t), bio-enhanced units optional	Hudson formula, K_D = 2 (extrapolated beyond 1:5); record of Section 6.1
Burial margin	0.5 to 1 m on crest level	seasonal bed change and settlement (Section 7.4)
Monitoring	seasonal bed and beach surveys before and after, with control sections; nourishment record	Section 5.3

Before design, the following are needed: (1) multibeam bathymetry of the site and seasonal bed surveys to test the 1 m observation; (2) a west-north-west wave climate giving the days with $H_0 \geq 0.6$ m; (3) a two-dimensional wave model or flume test including refraction and focusing; (4) calibration of the breaking criterion against field breaking depths; (5) seam and fabric tests if any geotextile is retained; and (6) before/after monitoring with controls. A Haifa reef designed this way would combine the features of the reefs that work (a large rock structure with a long, gentle ramp) with the advantage of a small tide, while accepting that it serves intermediate to advanced surfers and acts as a control point for the beach rather than as stand-alone protection.

8 Discussion

8.1 The author’s observations against the evidence

Table 6 sets the author’s eight observations (Section 1.3) against the evidence of Sections 3 to 7 and against the literature check of this review. Five are supported, at least in direction, and three partly; none is contradicted as a whole, and two parts are not supported (that “most tubes tear” at the break, and the sand trap as a source of sand).

Table 6. The author’s observations (a) to (h) tested against the evidence (verdicts by this review).

Observation	Verdict	Main evidence	Literature
(a) Reefs need volume; small reefs such as Pratte’s do not work	supported (descriptive)	volume ranks with outcome, ρ = 0.72 (p = 0.021, n = 10) stated and 0.93 (p = 0.005, n = 8) modelled; all reefs that do not work are 4,900 m³ or less, both built reefs that work 25,000 m³ or more; Burkitts (a reshaped natural reef) is the exception (Section 4.5)	consistent with Shand (2011, as cited in Mead & Borrero, 2011) and Borrero and Nelsen (2003); the threshold is new

Observation	Verdict	Main evidence	Literature
(b) Little measurable effect on sand; Gold Coast claims confounded by nourishment	supported where testable; Gold Coast cannot be decided	observed lee response 0.02 to 0.06 of the Black and Andrews (2001) prediction (Boscombe +3.4 m against 84 m); Palm Beach “retained” 550,000 m³ against about 470,000 m³ placed (Sections 5.3-5.4)	consistent with Ranasinghe and Turner (2006) and Blacka et al. (2013)
(c) Soft structures do not last in the breaking zone; geotubes are cheapest and support marine life	durability supported; cost partly; ecology untested	6 of 7 fabric reefs failed or were lost against 0 of 4 rock reefs (Fisher p = 0.015; confounded by date and budget); A$56 against A$457 per m³ to build, but renewal and removal add cost (Section 6)	durability consistent with Mead (as quoted in Geosynthetics Magazine, 2023); cost and ecology claims new
(d) Reef waves work on few days; Boscombe too steep, only for bodyboarders	supported; “only bodyboarders” partly	surfable 0 to 0.70 of the time even when a suitable swell is present; Pratte’s 1 to 5 breaking events a year; Boscombe 1:3 face collapses in every case; mostly, not only, bodyboarders (Sections 4.3, 4.6)	consistent with Mead and Borrero (2011) and Rendle (2016)

Observation	Verdict	Main evidence	Literature
(e) Wave power at the break lifts and drops the tube, so most tubes tear	partly	critical state just before breaking with simultaneous uplift (Bar & Drimer, 2023); sealed-base tubes 1.0 to 1.65 m high lift at 2.3 to 3.9 m depth; tearing not modelled, and reported causes are point damage, undersizing and construction faults (Section 6.3)	uplift consistent with Bar and Drimer (2023); tearing mechanism new and untested
(f) The depth gap alone does not decide; a long ramp gives a good wave, a sudden rise a too-hollow one	supported (direction)	same breaking depth (1.78 m) at Palm Beach and Boscombe, but ξ_b 0.56 against 3.00; depth gap ρ = 0.93, ramp length 0.85, crest depth 0.00 (Section 4.2)	ramp is a known component (Mead & Black, 2001a, as summarised by Blacka et al., 2013); the quantification is new
(g) Beginners need spilling waves that reefs do not make; difficulty set by peel angle; about 30 degrees for good surfers	supported	spilling breakers 0 to 29 % of on-reef cases; only one case (Borth, H_0 = 3 m, 61.9 degrees) reaches the 60 degrees of a learner; the 30 degrees matches the rating-7 limit of 29 degrees (Section 4.4)	consistent with Hutt et al. (2001); Walker's (1975) 30 degrees is a hydrodynamic minimum, not a skill class
(h) Israel: small tide gives consistency and smaller loads; mobile sand buries but could be trapped	partly	consistency supported (0.87 to 0.98 of the no-tide time kept) but the average does not rise; storms, not tide, set loads, and the Ashkelon North tubes tore; burial is a real risk, the sand trap supplies less than 4 % of one year's drift (Section 7)	Netanya tubes not confirmed (Ashkelon North instead; State Comptroller of Israel, 2020); bed change and sand trap new

Two cases deserve comment. The Pratte's testimony separates the two failures that (a) and (c) run together: the witnesses describe a wave that almost never broke, and one says the bags were intact when removed (Dan Harmon, 2026; grade D), whereas the removal report found fabric that "could be torn by hand" (Blacka et al., 2013, pp. 116-117). Either way the reef failed first as a wave, through its size and its site in the swell shadow of Catalina Island, which supports (a) more directly than (c). And Albany, the best wave in the sample, has a crest line rising at about 1:4 (Table 2) and is "intermediate-to-advanced on most days" (Mercer, 2025): a steep face can give a good but demanding wave when the reef is large and the swell long, so (f) describes the hollowness, not the value, of the wave.

8.2 Limitations and what would change the conclusions

The sample is small (8 to 13 reefs), and the outcome classes are this review's readings of sources that are mostly by the designers; every association is descriptive and confounded, because the large reefs are also the recent, better-funded rock reefs. No reef has a published wave climate at the structure, so the surfable fractions are shares of the time with a suitable swell, not days per year, and the reported surfable days are not comparable (Section 3.6). The geometry is idealised for six of the nine models, the incidence angle is assumed for five of eight reefs, and the wave model is one-dimensional linear shoaling with $\gamma = 0.78$ and no focusing; the Mead and Black (2001b) fit is extrapolated beyond 1:9 at Boscombe, Bunbury and Albany. Three key sources (Hutt et al., 2001; Mead & Black, 2001a; Black & Andrews, 2001) were used only as reproduced by others. The Haifa analysis rests on a synthetic profile and on unpublished observations of the author.

The conclusions would change with (1) measured surfable-day counts tied to a wave record at any reef; (2) a small reef that works, or a large rock reef whose wave fails, which would weaken (a); (3) a controlled before/after survey showing a reef-made salient at a site without nourishment, against (b); (4) a geotextile reef that survives a decade in the breaking zone without renewal, against (c); and (5) Haifa breaking depths that confirm the early breaking observed by the author (H_s/d of about 0.35 to 0.40), which would move the recommended crest from 0.6-1.0 m to 1.2-1.5 m below MSL.

8.3 Overall rating of the built reefs

To answer the question that motivates the review, which reefs have delivered what they promised, this review rated each reef on three criteria: (A) surf quality, the quality and consistency of the wave reported or computed; (B) safety and longevity, the structural record of the material and any hazard to users; and (C) sediment and storm protection, any measured effect on the beach or on storm damage. Each criterion is scored from 1 (very poor) to 5 (very good), with 3 for mixed or neutral evidence, including "no measurable effect"; ? means that no usable evidence exists, and * that a signal exists but cannot be attributed to the reef. The reefs are ranked by A, with B as tie-break; C does not affect the rank, because it is mostly unknown. The scores are this review's assessment, based on the evidence cited in Sections 3 to 6 and Appendix C, graded A (measured), B (engineering report or thesis), C (journalism or surfer media) and D (forum or single anecdote) (Table 7).

Table 7. Rating of the built surf reefs by this review (1 = very poor to 5 = very good; ? = no evidence; * = signal not attributable to the reef).

Rank	Reef	A surf	B safety, longevity	C sediment, storms	Reason (evidence)
1	Albany	5	2	?	best wave, rides of 100-120 m (Raised Water Research, n.d.-d); injuries, 18 rocks repositioned in 2026 (Sheil, 2026); too new for sediment data
2	Cables	3	4	3	rock, intact since 1999; surf-day claims conflict (Mead & Borrero, 2011; Blacka et al., 2013)
3	Burkitts	3	4	?	reshaped natural rock, no maintenance in nearly three decades; about 30 surf days a year is an estimate (Howells & O'Neal, 2024; Raised Water Research, n.d.-b)
4	Palm Beach	3	4	3*	rock, stable; wave only in E or S swell with H_s > 0.8 m, T_p > 7 s (Prenzler et al., 2023); nourishment confounds the shoreline (Section 5.4)

Rank	Reef	A surf	B safety, longevity	C sediment, storms	Reason (evidence)
5	Narrowneck	2	3	2	“rarely surfed” (EcoShape, n.d.); containers settled 1-2 m and were renewed in 2017-18 (Corbett et al., 2023); nourishment confounds
6	Borth	?	4	3*	rock, no damage in 2012-2024 imagery; no surf record (Blacka et al., 2013); +30 m bulge not attributable (Section 5.3)
7	Boscombe	2	1	2	too hollow, mostly bodyboarders (Rendle, 2016); a container lost and the reef closed in 2011
8	Kovalam	2	1	3*	good first season (Collins, 2010), then “most of the waves just roll over the reef” (Raised Water Research, n.d.-c); containers torn and sunk by 2013 (Blacka et al., 2013); beach gain only designer-reported
9	Mount Maunganui	2	1	1	hollow at first, later only in “a very confined set of conditions” (Mead & Borrero, 2011); deflated, removed 2014; council reported a scour hole and more rips (NZ Herald, 2014)
10	Pratte’s	1	1	2	too small; 1-5 breaking events a year (Borrero & Nelsen, 2003) and witnesses saw it break almost never (Dan Harmon, 2026, grade D); removed 2008-10
11	Opunake	1	1	?	no expected waves; bags failed, never completed (Reid & Sumner, 2021)
12	Bunbury Airwave	1	1	?	inflatable, tore during installation (ABC News, 2019)
-	Xala	?	?	?	no published information

Albany is the only reef that clearly delivers a high-quality wave, but with safety problems: reported injuries led to the repositioning of 18 rocks within a year of opening (Sheil, 2026). Hard rock reefs (Cables, Burkitts, Palm Beach, Borth) score best on B, with no structural failure; geotextile and inflatable reefs score worst on B, with six of the seven at the minimum. Criterion C is unknown or confounded for every reef: no score above 3 survives attribution, and the only negative evidence (Mount Maunganui) comes from a press report. The rating thus repeats the central result of this review: a surf reef that works is large, built of rock and given a long ramp, and its benefit is surfing rather than beach protection.

9 Conclusions

1. **Record.** Of 13 surf reefs built or reshaped since 1997, three work for surfing (Burkitts, Palm Beach, Albany), three work rarely, two are disputed, three do not work and two are unknown (Section 3). In this review’s rating (Section 8.3), only Albany clearly delivers a high-quality wave, and it has had safety problems; rock reefs score best on safety and longevity, geotextile and inflatable reefs worst, and the effect on sediment is unknown or confounded at every reef.
2. **Size.** The reefs that work are large: the three that do not work are 4,900 m³ or smaller, both built reefs that work 25,000 m³ or larger (Spearman $\rho = 0.72$ to 0.93; descriptive). Volume buys relief and ramp length.
3. **Ramp, not gap.** The crest depth decides which waves break; the ramp gradient decides how. Over similar depth gaps, a long 1:17 ramp gives a plunging, surfable wave and a 1:3 rise a collapsing one (Section 4.2), which explains the Boscombe wave, steep and suited mostly to bodyboarders and advanced surfers.
4. **Surfer level.** Refraction brings every reef arm to peel angles of 16 to 62 degrees at the break, suited to intermediate to professional surfers; no reef offers a beginner’s wave, which would need a spilling ramp of about 1:30 (Section 4.4).

5. **Sand.** The observed shoreline response is 0.02 to 0.06 of the standard prediction at the sites that can be tested; on the Gold Coast the reef effect cannot be separated from nourishment. A surf reef can serve as a control point within a nourishment scheme, not as stand-alone protection (Section 5).
6. **Materials.** Six of seven fabric reefs suffered major damage or loss within a few years, none of four rock reefs did (confounded by date and budget). Breaking-zone loads lift small or sealed tubes, but the tearing mechanism is unmodelled (Section 6).
7. **Haifa.** The small tide (spring range 0.4 m) makes a reef wave depend on wave height alone, not on the hour, but does not reduce storm loads. A Haifa reef should be rock, 100 m or longer, 15,000 to 25,000 m³ per 100 m, with a crest 0.6 to 1.0 m below MSL, a seaward ramp of 1:16 to 1:20 (never steeper than 1:12) and a 0.5 to 1 m margin for burial and settlement. Multibeam and seasonal bed surveys, a local wave climate, field calibration of the breaking depth and a two-dimensional wave model or flume test are needed before design, and before/after monitoring with control sections after construction (Section 7.5).

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Works cited only through a secondary source

These works are cited in the text as "as cited in" the listed secondary source and are not listed separately (APA 7, Section 8.6):

- Shand, T. D. (2011). *Making waves? A philosophical review of artificial surfing reef projects* (submitted to Shore and Beach); cited as quoted in Mead and Borrero (2011).
- Weppe et al. (2007), Mount Maunganui monitoring; cited as quoted in Blacka et al. (2013, p. 103).
- Geofabrics (2013), design life of geotextile containers; cited as quoted in Blacka et al. (2013, p. 35).

Appendix A. Methods and sources for the individual reefs

A.0 Common workflow

The same workflow was applied to all 13 reefs. It has six steps, and every step leaves a written record that can be traced from the numbers in the tables (Appendix C) back to the source.

1. Sources. For each reef the published design and monitoring papers, official reports, survey data sets, georeferenced aerial and satellite images, press articles and the pages of reef-specific websites were collected. Each item was recorded with its full citation, link, licence, access date, what it shows and how it was used. Every number in the dataset carries a source, the place where it was read (page, table, figure or file), the method used, and a status: sourced (read in a source), derived (computed in this study), assumed (an assumption, flagged A1, A2, and so on in the sections below) or unknown (Appendix C). Values from secondary compilations whose cited sources could not be fetched and confirmed were not used.

2. Plan shape. The outline of each reef was traced on the best available image, in a fixed order of preference: a measured survey (laser or multibeam) of the structure, then a design or construction drawing of the built version, then a photograph. Satellite images (Esri World Imagery and dated Esri Wayback releases) were georeferenced from their Web-Mercator tile bounds. Drawings and figures were scaled from a printed scale bar or a known dimension and placed by fitting to an image or a survey. Each outline was checked on a second, independent image or raster (overlap measured as intersection over union) and against the dimensions given in the text. Where more than one edge is defensible, the other outlines are kept as named, dated versions. Each outline is expressed in a local frame in metres (origin at the shoreline point nearest the centroid, x alongshore, y offshore), with the shoreline bearing from the same image. Confidence follows a fixed rubric: high if the structure is clearly visible on a georeferenced image or an as-built survey and the dimensions agree with the text to about 10 %; medium if the trace is on a scaled design drawing, only part of the structure is visible, or the dimensions differ by 10 to 25 %; low if no image of the structure exists and the shape is inferred from text or a sketch.

3. Three-dimensional reconstruction. A model was built only where the plan, the crest level and the seabed depth are each sourced. The seabed is taken from the best measured data (single-beam soundings, airborne lidar, multibeam read from a published figure, a coastal digital elevation model) and otherwise from a chart (Garmin Navionics SonarChart read in a web browser, datum tested; Garmin Navionics, n.d.) or from design contours. The reef surface is either the surveyed surface or a loft from the crest level, over the traced toe outline, down to the seabed; its volume is a mound envelope (voids included) and is not the fill volume. All heights are brought to mean sea level ($z = 0$ at MSL) through an explicit datum chain (Appendix C, note 1). Only local tidal planes of the nearest standard port are used. A model with an invented platform height or bed would look like data; where the three inputs are not all sourced (Burkitts, Opunake, Kovalam, Xala) no model was built.

4. Derived quantities. The depth profile was taken on the shore-normal transect through the shallowest crest cell. The depth gap is the seabed depth at the toe minus the crest depth, the ramp length is the horizontal toe-to-crest distance, and the mean slope is their ratio (definitions and sensitivities in Appendix C, note 2).

5. Documentation. For every reef the assumptions (numbered A1, A2, … with their effect on the result), the decisions (D1, D2, …, with who decided and why), the validation against independent values, the confidence per element and the unknowns, with what would resolve them, are written in a methods file next to the model. This appendix condenses those files.

6. Tools. Image analysis used threshold and morphological operations, normalised cross-correlation and phase correlation for image registration, distance transforms for lofted surfaces, and piecewise-cubic interpolation for seabed profiles (Python with NumPy, SciPy, Shapely, pyproj and PyMuPDF). The decisions taken by the author are marked D below (see Section 2.6 on the tools used). Readings of values from figures are by eye or by calibrated colour or tick scales, and carry the stated reading error.

Table A1 summarises the status of the 13 reefs. Three-dimensional previews of the models are shown in Appendix B.

Table A1. Plan outline, data used for the reconstruction, and confidence. Plan confidence follows the rubric in step 2. “3D” gives the overall confidence of the three-dimensional model.

#	Reef	Default plan outline	Plan	Seabed	Crest level	3D
A.1	Burkitts	visible platform, Esri 5 Sep 2025 (stage, not the reef)	low	none	none	not built
A.2	Cables	design plan of Lyons (1992) fitted to the 1999 soundings	high	single-beam survey, Mar 1998	surveyed rock surface, Nov 1999	medium
A.3	Narrowneck	visible container field, Esri 8 Aug 2020	medium	2017 multibeam read from a figure; Navionics	measured crest -2.2 m AHD (2018)	medium
A.4	Pratte’s	Phase I design layout, Skelly Engineering	medium	NOAA 1/3 arc-second DEM	apex -6 ft MLLW, constant thickness	low
A.5	Mount Maunganui	-2.0 m CD contour of the 2013 multibeam	medium	Navionics 2026, bed level assumed	-0.9 m CD (council report)	medium
A.6	Opunake	consent occupation area only (not the reef)	low	none	design text only	not built
A.7	Boscombe	bag field on Esri image of 28 Sep 2011	medium	DGPS bathymetry, Apr 2011	designers’ +0.5 m ACD	medium
A.8	Kovalam	photo trace at assumed camera height	low	none	none	not built

#	Reef	Default plan outline	Plan	Seabed	Crest level	3D
A.9	Borth	laser-survey edge, LiDAR 19 Mar 2022	high	design bed levels, contour figure, EMODnet	LiDAR and drawings	medium
A.10	Palm Beach	toe outline of the City asset polygon and aerial	high	Navionics	-1.5 m MSL (four sources)	medium
A.11	Bunbury	circle of 12 m on a drone photograph	medium	WA lidar 2009	height 1.6-2.0 m (text)	medium
A.12	Albany	Bluecoast design outline	high	2015 design contours	-1.0 m AHD (City design)	low
A.13	Xala	none	not traceable	none	none	not built

A.1 Burkitts Reef, Bargara, Queensland (1997)

Sources. Burkitts Reef is a natural basalt headland platform at the south end of Bargara Beach. In February 1997 about 300 m^3 of boulders (at least 12 large ones) were moved with an excavator at a cost of about AUD 10,000 (Rendle, 2016, Table 2.1; Howells & O'Neal, 2024; Raised Water Research, n.d.-b). No source gives a length, width, crest level or depth. The work zone of a later phase is shown on an annotated oblique aerial photograph (Jackson & Corbett, 2007, Fig. 1). The only georeferenced image is Esri World Imagery of 5 September 2025 (Esri et al., n.d.). Tidal planes of the nearest standard port, Burnett Heads, are from Maritime Safety Queensland (2026; MHWS 2.94 m, MSL 1.78 m, MLWS 0.61 m above LAT).

Plan shape. The visible platform was traced on the 2025 image (31,485 m^2, a lower bound because the image frame cuts it). The later work zone was georeferenced with three control points on an Esri Wayback image of 31 July 2016 (an ellipse of about 1,750 m^2, accuracy 20-30 m). The traced outline is the platform on which the reef sits and not the reef, because no source locates the 1997 works; the outline is therefore not used in the analysis (confidence low). The assumptions A1-A8 of the shape file are that the platform is the stage, that its edge is the breaker line at an unknown tide, and that nothing has changed since 1997.

3D reconstruction. Not built: crest level, platform height and seabed depth are all missing (the depths quoted for the dive site are site-wide ranges, not a profile).

Unknown and what would resolve it. The reshaped zone, the crest depth, the seabed and the tide at the time of the image; dated low-tide imagery with tide tables, a chart reading or a local account would resolve them.

A.2 Cables Reef, Perth, Western Australia (1999)

Sources. The design plan of Lyons (1992) as reproduced in Blacka et al. (2013, Fig. A.2) gives the V-shaped outline with design levels (RL); the design study (Pattiaratchi, 1999) and the copies of the drawings on Raised Water Research (2019b) give the 45-degree peel angle, the 1:20 ramp and the proposed extension of the southern arm. The measured data are the single-beam soundings of the Western Australian Department of Transport (2020): a pre-construction survey ending 30 March 1998 (14,303 soundings in the window), and surveys of 22 November 1999 and 1 February 2002 (heights in m AHD). The Australian Height Datum (AHD) is defined by Roelse et al. (1975). Tidal range and mean level are from the Fremantle gauge about 4 km south of the reef.

Plan shape. The silhouette of the drawn structure (111 vertices) was scaled from four printed dimensions (6.51 px per metre, sd 0.05). Its position and rotation (9.74 degrees) were found by maximising the share of 1999 rock soundings inside the outline minus the share of bed soundings inside it: 91.0 % of rock and 18.5 % of bed soundings lie inside. With the scale also free, the optimum scale was within 2 % of the drawing scale. The outline coincides with the dark rock patch on four Esri images. Its size (121 by 76 m, 4,049 m^2) differs from the text (140 by 70 m) by -13 % and +9 % (confidence high; extension version 4,355 m^2, unconfirmed).

3D reconstruction. The seabed is the 1998 survey averaged per metre and interpolated, outside the reef the DoT composite grid shifted by +0.085 m, smoothed with a 2 m Gaussian filter (bed at the anchor -3.68 m AHD, accuracy 0.3 m). The default rock surface is the thickness (sounding minus modelled bed) at the November 1999 soundings inside the footprint and a 6 m margin, interpolated linearly to a 0.5 m raster, with negative values set to zero and a 1 m Gaussian smoothing (the rule that a sounding more than 0.6 m above the 1998 bed is rock was used only to classify soundings for the georeference fit); volumes are 4,671 m^3 (1999) and 5,219 m^3 (2002) (accuracy 1,000 m^3), against 3,500-5,500 m^3 in the literature. Design-surface versions loft the printed contours.

Assumptions and decisions. A1: the design RL is Fremantle low water datum, so z_{AHD} = RL - 0.716 m. A12: AHD is taken as MSL (local MSL 1992-2011 is 0.057 m higher). A5: side slopes of the design versions are linear over the drawn bands. A6: the extension is a pyramid scaled to 320 m^3 (unconfirmed). A14-A17: the 1998 bed is the bed under the reef; soundings are rock-top depths, and interpolation between lines of about 5 m smooths boulder-scale relief. A7: high and low water are taken as MSL +-0.4 m.

Confidence. Plan high; seabed medium-high; surveyed rock surface medium; datum medium-high; tides low-medium (range and MSL measured, HAT and LAT not found); overall medium. The surveyed rock lies 0.13-0.22 m above the design ramp, which is not a datum effect.

Unknown. The extension, the rock surface after 2002 (a 2009 critique by A. Pitt, quoted by Raised Water Research, 2019b, puts the crest about 1.5 m below the lowest tide), HAT and LAT, and the sign of a 0.04 m Perth-zone height difference.

A.3 Narrowneck Reef, Gold Coast, Queensland (1999-2000, renewed 2017-18)

Sources. The monitoring papers (Jackson et al., 2007; Jackson et al., 2012), the renewal paper (Corbett et al., 2023), the sediment-pathway study (Vieira da Silva et al., 2021), the decade review (Mead & Borrero, 2011) and the NSW review (Blacka et al., 2013). The plan is traced on Esri Wayback release 9812 of 8 August 2020 (Esri et al., n.d.). The seabed is read from Fig. 3 of Corbett et al. (2023), a multibeam map of 2017 before the renewal, and from Garmin Navionics SonarChart labels. Tidal planes are from Maritime Safety Queensland (2026) for the Gold Coast Seaway; the asset polygon of the City of Gold Coast (n.d.; an envelope of 34,565 m^2) was not used for the outline.

Plan shape. The visible containers were extracted by a threshold on the green channel, closing of 3.7 m and hand-checked polygons (7 polygons, 6,239 m^2), georeferenced from the tile bounds (0.264 m per pixel; edge accuracy 1 m; waterline 15 m). Independent checks: a second tile set of November 2022 agrees to 0.2 m (phase correlation), images of 2016 and 2019 to 3-4 m by eye, and Navionics shoals to 7-25 m. The footprint is a lower bound because the deeper toe is not visible. Four more outlines are selectable: the Option 2 renewal design of 2017 (13,941 m^2), the revised design of 2004 (13,631 m^2), the relief zone of the 2011 survey (4,922 m^2) and the Navionics 4.0 m shoals (537 m^2). **Decision D7 (author, 10 October 2026):** the default is the 2020 photograph trace, because it is the only evidence of the state actually built (Option 2 was built only in part on the south reef) and no survey or drawing of the renewed reef exists.

3D reconstruction. The crest is the measured -2.2 m AHD of the 2018 multibeam (Vieira da Silva et al., 2021). The multibeam map of Corbett et al. (2023, Fig. 3) was converted from colour to depth with a linear fit through four tick labels of its colour bar (depth accuracy 0.4 m, position 10 m); the bed is a grey-scale opening (radius 4.65 m) and 6-pixel Gaussian smoothing of that raster, blended into an alongshore-uniform profile (seaward labels 7-10 m from Navionics; shoreward part assumed). The reef surface is the design surface (linear in distance rank between the -2.5 m crest loop and the -6 m envelope) lowered to the measured crest. Volumes are mound envelopes.

Assumptions and decisions. A1: the figure depths are m AHD. A3: the colour scale is linear. A4: the design surface between the contours is linear (offset +0.26 m, sd 0.66 m against 53 multibeam blocks). A6: Navionics depths are below LAT. A7: the nearshore bed (0-190 m) is a smooth monotone profile. A10: the datum link AHD = LAT + 0.760 m of Maritime Safety Queensland is used instead of the +1.00 m implied by the papers; this moves reef and bed by 0.24 m relative to the tidal planes (crest 1.44 m below LAT, or 1.20 m on the papers' rule) but does not change the geometry.

Confidence. Plan medium; crest medium-high; seabed in the window medium (seaward low-medium, nearshore low); reef height and slopes medium-low; tides medium-high; volume low-medium; overall medium.

Unknown. The renewed plan and survey (City of Gold Coast drawings, 2018-19 multibeam), the datum of the original multibeam, the nearshore bed, and the container-level shape.

A.4 Pratte's Reef, El Segundo, California (2000-2001)

Sources. The design plan of Skelly Engineering (c. 2000), reproduced as Fig. 4 of Borrero and Nelsen (2003); the monitoring paper (Borrero & Nelsen, 2003) for the crest datum (apex "approximately -6 ft MLLW"), bag size, dates and the survey window of its Fig. 9; the NSW review (Blacka et al., 2013); the NOAA Santa Monica 1/3 arc-second coastal digital elevation model (NOAA National Geophysical Data Center, 2010) for the seabed; Navionics depth labels as a check; tidal datums from the NOAA Center for Operational Oceanographic Products and Services (NOAA CO-OPS, 2026; Santa Monica, 11.7 km away, epoch 1983-2001).

Plan shape. The solid-outline bags of the design drawing (110 bags, 76 vertices) were traced and scaled from the printed bar (5.155 px per metre; agrees with the known bag size to 3-6 %); two reproductions agree in area to 1 %. The result (435 m^2, 58.0 by 28.6 m) is a design layout, not an as-built plan; the 30 optional bags are excluded. Position: the centre of the survey window of the paper's Fig. 9 (33.92058 N, 118.43335 W, accuracy 150 m).

3D reconstruction. The Phase I reef follows the DEM seabed at a constant thickness of 1.349 m (one 1.2 m bag course) so that the apex face lies at -2.678 m MSL (-6 ft MLLW, $z_{MSL} = z_{MLLW} - 0.849$ m). The bed is the mean of 31 alongshore DEM profiles, checked against Navionics labels and the paper's profiles (agreement 0.3 m). Phase II and the removal in 2008 and 2010 are not modelled.

Assumptions. A2: the reef centroid is 91.44 m from the MSL waterline (+-25 m moves the bed by 0.78 m). A4: the 6 ft depth applies at the apex face, and the top follows the bed at constant thickness. A5: the bag-stack side slope is 1:1 (0.5-2.5), which is also the **assumed ramp and slope** in Appendix C. A9: the footprint is the design layout. A10: no settlement, scour or burial is modelled, although the real reef sank and was buried. A13: the position is the survey-window centre.

Confidence. Plan medium; datums high; seabed medium; stack height and slope low; overall **low** (design footprint, inferred position, assumed height and slope, and a height uncertainty of 0.85 m against a reef height of 1.35 m).

Unknown. The as-built footprint and bag layout, the crest elevation along the reef, the number of courses, and the true position.

A.5 Mount Maunganui Reef, Bay of Plenty, New Zealand (2005-2008)

Sources. The council assessment of management options (Dahm & Gibberd, 2013), whose Fig. 3 is a multibeam survey of 18 July 2013, and Fig. 4 a survey of January 2007; the design papers (Mead & Black, 1999; Mead et al., 2007); Mead and Borrero (2011); the NSW review (Blacka et al., 2013); an ASR image of the completed reef in colour-coded depth (undated, after August 2008, datum not stated; ASR Ltd, via Raised Water Research, 2019c); aerial photographs of 2010-11 (Land Information New Zealand, 2010-2011) and Esri Wayback release 3630 of 15 January 2011; Navionics SonarChart of 2026; tide levels from Land Information New Zealand (LINZ, n.d.).

Plan shape. The default is the -2.0 m chart-datum (CD) contour of the 2013 survey (1,071 m^2), digitised on the NZ-grid ticks of Fig. 3 (5.45 px per metre) and confirmed on a second raster (area 1,074 m^2, intersection over union 0.93) and on the 0.125 m aerial (position within 2 m). The second version is the as-built toe outline of the ASR image (1,424 m^2, scale +-6 %), corroborated by the as-built survey in Fig. 3 of Mead and Borrero (2011). The 2013 outline is five years after completion (apex bags deflated, base bags buried) and is a lower bound.

3D reconstruction. The reef is lofted from the traced outline to a crest at -0.9 m CD (z = -2.03 m relative to the present MSL, with MSL = CD + 1.13 m) over the 2026 Navionics seabed shifted to an as-built bed level of -3.6 m CD at the reef centre. The flank is 1:1 (default) or a linear loft from the toe (second version); the inner profile is the median rise curve of the 2013 digital elevation model scaled to the crest.

Assumptions and decisions. A6 and **decision D1 (author, 7 October 2026): the as-built bed at the reef is -3.6 m CD**, an assumption, because no source states the 2008-09 bed; every later source gives a bed 0.8-1.2 m shallower, and the stated built volume of 2,800 m^3 is reproduced only with a bed of -3.51 to -3.66 m CD. A bed of -2.8 m CD would give a height of 1.9 m instead of 2.7 m (sensitivity: +-0.5 m is +-22 % in volume). A3: the crest is -0.9 m CD on every bag. A5: the 2026 seabed shape stands for 2008. A7: flank 1:1 (a run of 0.25-2 changes the volume from 2,567 to 3,421 m^3). The scour hole behind the reef (to -4.9 m CD in 2013) is not modelled.

Confidence. Plan medium (default) and medium-low (toe version); crest medium-high; tides and datums high; seabed shape medium; **bed level and height low (assumed)**; volume low-medium; overall medium.

Unknown. The 2008-09 bed at the reef, the datum of the ASR image, a section drawing, and the as-built footprint (historical imagery of 2008-10 could provide it).

A.6 Opunake Reef, South Taranaki, New Zealand (2006-2009)

Sources. The monitoring report of the regional council (Taranaki Regional Council, 2009), with the coordinates of Coastal Permit 6376-1 (31 May 2005) in its Appendix I and an aerial photograph in its Fig. 1; the NSW review (Blacka et al., 2013, Tables 4.2-4.4); Mead and Borrero (2011); a magazine article (Moores, 2006); a newspaper report (The Daily News, 2004); a news feature (Reid & Sumner, 2021); and aerial photographs of 2011-12 (Land Information New Zealand, 2011-2012), which show no structure.

Plan shape. The four published corner coordinates (a local circuit grid, corrected for one misprinted easting) form a rectangle of 189 by 94 m (17,726 m^2), centred at 39.45943 S, 173.85163 E, georeferenced against Fig. 1 (residual 0.3 m, independent check 0.7 m). This is the **legal occupation area, not the reef**. The reef itself is described only by its design size (99 by 30 by 3.5 m, 4,900 m^3; Taranaki Regional Council, 2009, p. 4). Where the reef lies inside the area is unknown (+-47 m along the axis, +-32 m across); the 99 by 30 m rectangle was assumed to be centred and parallel (A7, A8) and is not shown. The earlier size “90 by 20 m” is the 2004 concept quoted in The Daily News (2004) and repeated by later articles.

3D reconstruction. Not built: no bag layout, no per-bag crest levels and no surveyed seabed exist. Depths are known only below chart datum (LAT): the shallowest contour is at 0.0 m and the seabed is 1.8-3.0 m below chart datum (Taranaki Regional Council, 2009, pp. 3-4); an MSL conversion needs the LINZ standard-port levels and was not made. The as-built survey required by the consent within 90 days of completion is held by the council.

Confidence. Position of the occupation area medium-high; reef shape low (envelope only).

A.7 Boscombe Surf Reef, Dorset, United Kingdom (2008-2009)

Sources. The design paper (Mead et al., 2010), the evaluation of the damaged reef with a DGPS bathymetry of April 2011 in its Fig. 9 (Rendle & Davidson, 2012), the doctoral thesis (Rendle, 2016), Mead and Borrero (2011), the NSW review (Blacka et al., 2013), Esri Wayback release 10 of 28 September 2011, beach profiles of 20 April 2010 (Channel Coastal Observatory, 2026b), the chart-datum offset of Bournemouth (National Tidal and Sea Level Facility, n.d.), and Garmin Navionics for validation.

Plan shape. The bag field is clearly visible on the 2011 image and was traced (4,042 m^2, 88.5 by 94.3 m), agreeing with the April 2011 survey zones and with the designers’ own plan. The area is about 40 % of the roughly 1 ha repeated in the literature (confidence medium). The outline is of the state after the damage of March-April 2011; the 2009 edge may differ by a few metres on the damaged offshore half.

3D reconstruction. The as-built state of 2009 is an idealised two-layer loft over the outline: a flat crest at the designers' +0.5 m above chart datum, 1:3 flanks, and a 1 m relief at the outline. The seabed is read from Fig. 9 (accuracy 0.2 m), with the beach from the measured 2010 profiles, and spline or EMODnet slopes beyond the survey. Heights are referred to MSL with $z_{MSL} = z_{ACD}$ - 1.40 m; tide levels are from the gauge table of the design paper. Sixteen Navionics soundings agree with the model seabed (mean difference +0.06 m, sd 0.49 m), and its shoal areas match the survey zones to 1-2 %.

Assumptions. A3: the zero of the survey plot is chart datum (if it were Ordnance Datum all absolute heights shift by 1.4 m, with relief and volume unchanged). A6: the built crest is flat at +0.5 m ACD (+-0.3 m is +-4 % in volume). A7: flanks are 1:3 (1:2 or 1:4 change the volume by +8 % and -7 %); this flank is the **assumed slope** behind the ramp of Appendix C. A8: the seabed under the reef is a spline of its surroundings. A12: the 54 containers are not modelled individually.

Confidence. Plan medium; seabed medium-high (measured, +-0.2 m); crest medium (designer value, checked against October 2009 profiles); datum assumption strongly supported but not stated by the authors; overall medium.

Unknown. The datum of the Fig. 9 plot (to be confirmed from the thesis or from survey metadata), the as-built plan (a 2009-10 aerial image), and the crest height along the whole reef.

A.8 Kovalam Reef, Kerala, India (2009-2010)

Sources. The ASR Ltd oblique aerial photograph of the completed reef from the lighthouse end (reproduced in Blacka et al., 2013, Fig. 4.5, where it is credited to ASR Ltd), a second oblique photograph with a hand-drawn break line (Raised Water Research, n.d.-c), Esri Wayback release 10 of 28 February 2011 for the shoreline, the NSW review tables (Blacka et al., 2013, Tables 4.2-4.4, pp. 66-70), Mead and Borrero (2011, Section 3.1.7) and a report by a surfer (Collins, 2010).

Plan shape. The five or six sand-container strips were traced in photograph pixels. Because the photograph has no scale and the camera height is not fixed, a camera model was fitted to the 2011 swash line (residual 2.2-2.8 px for heights of 30-80 m). The metric outline is therefore a family: 640 m^2 (35 m), 1,216 m^2 (60 m, the default) and 1,681 m^2 (80 m). The alongshore length agrees with a secondary table (65 m), but the cross-shore size does not (41 m against 85 m). Confidence low; the shape is hidden in the survey figures.

3D reconstruction. Not built: the sources give only a crest submergence range (0.2-3.4 m, datum not stated, not tied to a position), a water depth range of 1.3-3.7 m and a tide range of about 1 m, with no profile, contour or datum, and the reef later sank, so no as-built survey exists.

Unknown. The true size and position, crest and seabed depth, and tidal planes. A navigation-chart reading, historical commercial imagery for February 2010 to 2012, or the designers' original design report would resolve these.

A.9 Borth reef, Ceredigion, Wales (2011-2012)

Sources. The Phase 1 construction drawings 9V5090/1020-1023 and 1001, rev C1 of January 2011 (Haskoning UK, 2011), with plan rings, 15 setting-out points on the Ordnance Survey grid, crest levels, layer thicknesses and slopes; the laser survey (Welsh Government, 2022; 1 m DSM and DTM, flown 19 March 2022); Esri imagery of 2012-2024; the physical-modelling paper with the pre-construction contours (Rigden et al., 2013, Fig. 2); the tide table of the shoreline management plan (Royal HaskoningDHV, 2011); EMODnet bathymetry (EMODnet Bathymetry Consortium, 2024); a trade-press article for the rock quantities (Hansford, 2011); the national heritage record (Royal Commission on the Ancient and Historical Monuments of Wales [RCAHMW], 2020).

Plan shape. Two mounds, a northern boot-shaped surf reef and a southern oval **shore-parallel breakwater** (not a surf reef), are bare rock on four dated Esri images (2012-2024). The outlines agree with the georeferenced design plan in position (0.3-4 m), orientation (1-3 degrees) and length (-4 % north, +0.4 % south). Three outlines are selectable: the **laser-survey edge (default, chosen by the author on 6 October 2026; 8,675 m^2)**, the design rock-layer foot (6,989 m^2) and the visible rock edge in the photograph of 17 September 2024 (5,743 m^2, used for the plan area in Appendix C). The OS-grid to local-frame transformation has a residual of 0.005 m; the absolute position is limited by the datum transformation (2-5 m).

3D reconstruction. The surface is the 1 m LiDAR surface above the exposed level (about -2.3 mODN) plus the design toe berm and blanket of the sections below it. The seabed is the design bed level under the rock (about -4.0 mODN, "bed levels vary"), the hand-read -3.0 and -4.0 mODN contours of the physical-modelling figure around it, and an EMODnet gradient of -1.1 % beyond the contours. The model rock volume is 30,119 m^3 (51,200 t at an assumed 1.7 t m^{-3}, +-15 %). Heights: $z_{MSL} = z_{ODN}$ - 0.31 m; the tide slider stops at MHWS (+2.56 mODN), because HAT of Aberystwyth was not found at a primary source.

Assumptions and decisions. A2: MSL = +0.31 mODN, the mean of the four tidal planes (+-0.15 m). A3: LAT = -2.44 mODN. A4: the LiDAR datum is ODN. D6 (7 October 2026): the -4.0 mODN contour of the physical-modelling figure became a control after the seabed seaward of the reef was found to be 0.89 m deeper than that contour. Beyond y = 335 m the model is 0.7-1.2 m deeper than the EMODnet cells.

Confidence. Plan high; crest levels high (design drawing, LiDAR and section drawings agree within 0.3 m); tide offsets medium; seabed medium-low (the bed level under the rock is the first-order unknown for height and volume); overall medium.

Unknown. The bed under the rock (no survey), HAT, the LiDAR datum (+-0.15 m on the crest), and any reef-only rock quantity.

A.10 Palm Beach Reef, Gold Coast, Queensland (2019)

Sources. The asset polygon of the City of Gold Coast (City of Gold Coast, n.d.; boulder reef, 160 by 80 m, 11,942 m^2), Esri World Imagery of 1 December 2025, a vertical Nearmap aerial with survey contour lines and a 0-25-50 m scale bar (Bluecoast Consulting Engineers, n.d.), the concept design (Mortensen et al., 2015), the project paper with the final multibeam survey as an oblique image (Hunt et al., 2022, Fig. 4), Garmin Navionics SonarChart, and tidal planes of Maritime Safety Queensland (2026).

Plan shape. The outer contour of the aerial (toe outline) was traced and placed by the north arrow and scale bar; it matches the City polygon to 0.55 m (11,972 against 11,942 m^2) and an independent drawing of the same ICCE paper (about 45 degrees to the shore). The crest slot (59.2 by 5.9 m) is the innermost contour. Confidence high.

3D reconstruction. The ten unlabelled contour lines of the aerial were taken as the surveyed rock surface at an assumed interval of 0.5 m (a Navionics-derived interval of 0.46 m supports this), and the seabed as a smooth surface through the Navionics iso-lines outside 25 m of the reef and the toe elevations. The crest is at -1.5 m MSL (four sources). The two inferences agree to 0.27 m on average (sd 0.55 m, 427 points); no surveyed depth value was read.

Assumptions. A3: the contour interval is 0.5 m. A5: Navionics depths are below LAT and not tide corrected (LAT tested best, MSL rejected). A6: the natural seabed is smooth between iso-lines. A8: one rock surface with voids included. A12: the aerial lines show the as-built surface. A13: the Navionics seabed represents the 2019 bed. A10 and A11: the tidal planes of the Gold Coast Seaway apply (20 minutes earlier), with AHD 0.12 m below MSL.

Confidence. Plan high; crest level high; seabed and reef height +-0.5 m; overall medium. The City's asset register (height 5 m, volume 33,000 m^3, units unknown) describes a reef 20-33 % larger than the model (crest height 3.9 m, envelope about 22,000 m^3); the discrepancy is unresolved.

Unknown. A surveyed as-built seabed or crest value, the datum of Navionics, the interval and date of the aerial contours, and the units of the register entries.

A.11 Bunbury Airwave, Western Australia (December 2019)

Sources. A news report (ABC News, 2019), the developer's account in the local press (Bunbury Mail, 2022), a spoken interview with the founder (T. Bottegal, Airwave founder, pers. comm., 8 October 2026; an interested party), a drone photograph (Raised Water Research, 2019a) and a pre-installation plan graphic (Raised Water Research, n.d.-a), a video frame of the installation (ABC News, 2019), the airborne lidar of 2009 (Western Australia Department of Transport, 2009; 10 m cells, vertical datum AHD), Garmin Navionics SonarChart contours counted in a browser, and the tidal planes of the Port of Bunbury (GHD, 2021).

Plan shape. The bladder is a circle of 12 m diameter, traced with 29 vertices on a near-nadir drone photograph (112 m^2); a second video frame and the plan graphic agree. Neither image has a scale or a georeference, so the size comes from the stated diameter and the position (centre 37.5 m from the waterline, text range 30-45 m) from the text. The coordinates are an estimated centre (assumed). Confidence medium.

3D reconstruction. A symmetric spherical cap of height 2.0 m (range 1.6-2.0 m, volume 93-117 m^3) stands in for the real surface, on the lidar seabed (bilinear, resampled to 5 m). Lidar and Navionics agree (RMS 0.32 m once Navionics depths are read below LAT). The seabed in the text ("about 1 m under the surface at low tide") and the lidar agree to 0.22-0.62 m.

Assumptions. A2: the 2009 0 m AHD contour is the zero of "distance offshore" (+-10 m is +-0.41 m of seabed depth). A4: height 2.0 m. A5: spherical cap. A6: "low tide" is MLLW. A8: the 2009 seabed is unchanged in 2019 (sd 0.30 m, a judgement). A10: the lidar datum is AHD and its uncertainty layer (0.47 m) is one standard deviation. A11: Navionics datum is LAT with a 0.5 m interval.

Confidence. Seabed sourced and independently supported; tides sourced and cross-checked (0.03 m); surface shape assumed; overall medium. The verdict "failed" rests on all dated 2019 reports; the developer's page dates the trial to 2018 and calls it not a failure, a claim contradicted by the dated reports.

Unknown. The as-installed surface (which flank was steep, fill level, the torn quadrant), the exact position, seabed change since 2009, and any rideable-wave record.

A.12 Southern Ocean Surf Reef, Albany, Western Australia (2025)

Sources. Esri World Imagery of 20 January 2026 (the reef is a dark wedge, absent on 16 January 2025) and Esri Wayback release 13192 (Esri et al., n.d.); the Bluecoast final-design plan with contours, via Raised Water Research (n.d.-d), and the City of Albany design table (City of Albany, n.d.); the Option B plan with depth contours of Royal HaskoningDHV (2015, as reproduced by Raised Water Research, n.d.-d) for the seabed; the environmental impact assessment (BMT Western Australia, 2018) for soundings; tide predictions for Albany (Western Australia Department of Transport & Bureau of Meteorology, 2024); and construction reports (Sheil, 2026; Mercer, 2025).

Plan shape. The design outer contour (13,104 m^2, 127 by 153 m) was scaled from its bar and placed on the 2026 image; the satellite-visible rock patch (10,918 m^2) is the second version. The two outlines overlap by 81 %, with orientations within 3 degrees, and the length agrees with the City's 165 m (design plan 165 m, satellite outline 157 m). Confidence high. The default is the design outline, the volume of which (34,000 m^3 on 2017 bathymetry) is stated by the City.

3D reconstruction. The seabed is one shore-normal profile read from the colour contours of the Option B plan along eight transects (the plan was georeferenced on four control points, residual 3-6 m; nodes -1 m at 36 m to -8 m at 330 m from the beach-face line), assumed alongshore uniform. The reef is lofted between a crest ridge, offset 9 m inside the shoreward edge and at -1.0 m AHD for the first 15 m, falling to -2.0 m at 40 m, and the toe outline: $z = z_c + (z_b - z_c)(d_c/(d_c + d_t))^N$, with slopes limited to 1:2. The exponent N = 2.73 is fitted so that the design outline holds 34,000 m^3, so the volume is not an independent check. Rock mass uses 1.847 t m^{-3} (62,800 t / 34,000 m^3).

Assumptions. A2: AHD is MSL (LAT is 0.74 m below AHD, from the mean of 1,043 predicted high and low waters of 2024). A4: the seabed is alongshore uniform. A6 and A7: the crest ridge offset and the crest levels (the City gives -1.0 m "shallowest" and 110 m at -2 m, but not where). A9: N is fitted. A10: bulk density. The datum of the 2015 contours is not stated and is taken as AHD; it conflicts with a 2018 sounding by 1.6 m near the shore.

Confidence. Plan high; crest medium-low (design values, no as-built survey); seabed low; pile height and slopes low (height known to about +-1 m); tides low-medium; overall **low**.

Unknown. The Bluecoast detailed design report with the 2017 bathymetry, the completion survey of 2025, and the rock movements of May 2025 and February 2026 (not modelled). Depth readings, or a survey by the Department of Transport (Middleton Bay), would fix seabed and heights.

A.13 Xala Reef, Jalisco, Mexico (2025-2026)

Sources. An announcement by an industry website on 31 March 2026 (Indelicato, 2026), reporting a boulder reef designed by S. B. Mortensen (DHI Seaport), and a feature reporting "3,000 truckloads" of quarried boulders (Romeyn, 2025). The two images available show a surfer and a drone shot of a peeling wave; the rock reef is submerged and nothing is measurable.

Plan shape and 3D. Not traced and not modelled. No image of the structure, no scaled drawing, no dimension and no coordinate exist (the location was withheld until a reveal event billed for 16 April 2026), and an area pin found by the search is explicitly not the reef. A text-derived envelope of 160 by 80 m, placed 230-310 m offshore by analogy to Palm Beach Reef, exists in the project files but is not a measurement and was not used.

Unknown. All geometry, cost, contractor, construction dates (2025 or 2026) and performance.

Appendix B. Three-dimensional reconstructions

Nine of the 13 reefs had a sourced plan, crest level and seabed depth, and were reconstructed as three-dimensional models of the reef, the surrounding seabed and the local water levels (Section 2.3). The models show the as-built state; later changes such as settlement, renewal or removal are described in Section 3 and Appendix A, not drawn. All heights are relative to mean sea level through the datum chain of each reef (Appendix C, note 1), and only the tidal planes of the nearest standard port are used. The depth profiles of Figure 1, the computations of Sections 4 and 5 and the volumes marked "model" in Table 2 are taken from these models. Figure B1 shows the default version of each; the inputs, sources, assumptions and confidence of every model are listed per reef in Appendix A.

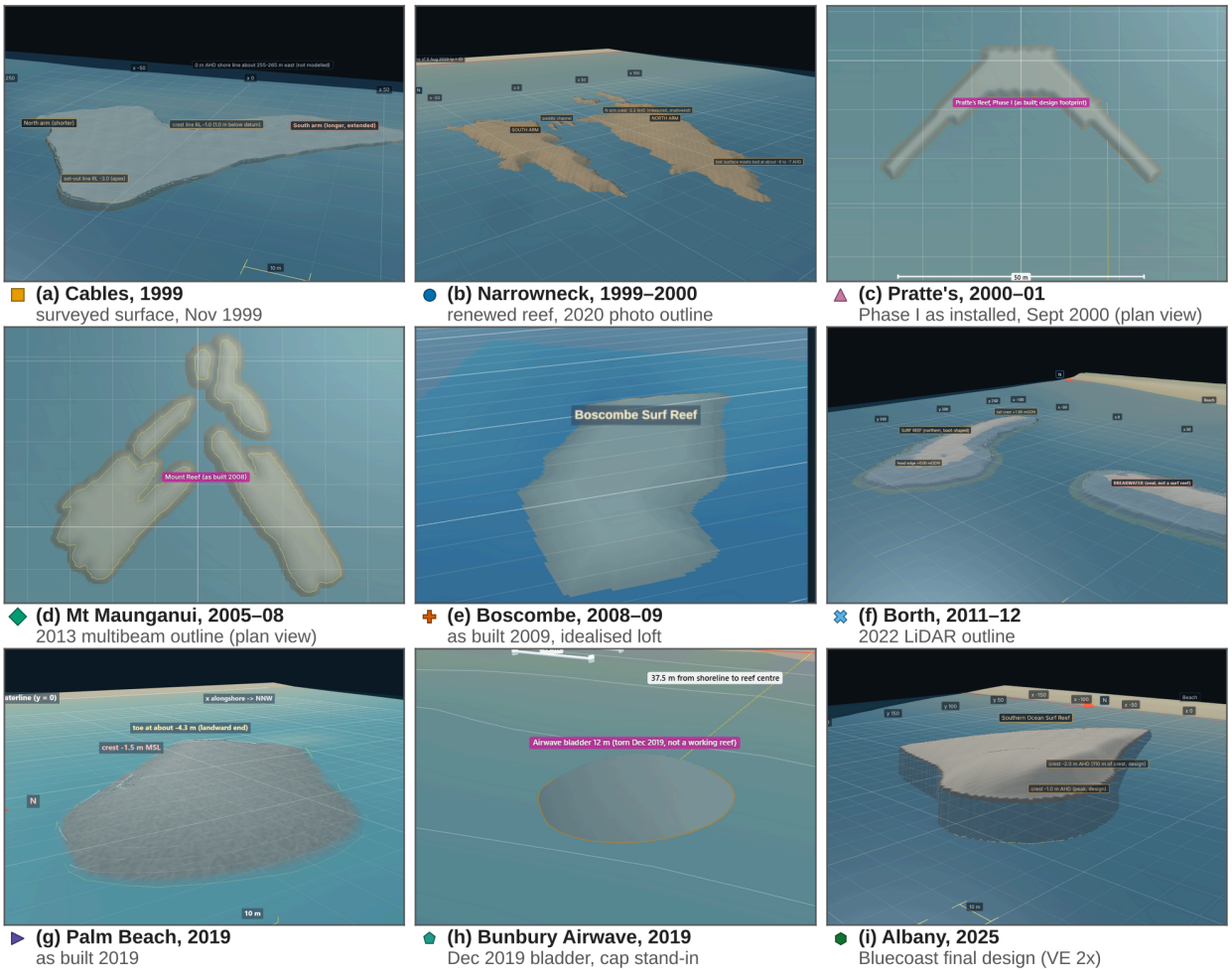


Figure B1. Three-dimensional reconstructions of the nine modelled reefs (default version of each model; screenshots of the interactive viewer, panels cropped away). These are reconstructions, not photographs or surveys: each combines a traced or designed outline, a surveyed or inferred seabed, crest levels from design or survey documents and the local tidal planes, and its confidence differs by reef; the inputs, sources and assumptions of every model are listed per reef in Appendix A. Tiles are not at a common scale (use the scale bars and labels inside each tile); views are oblique except (c) and (d), which are plan views because the oblique preview was cut by the viewer panels; vertical scale is true except (i) with 2x exaggeration.

Appendix C. Reef data table

This appendix reproduces the master data table of the review in condensed form: 13 surf reefs, ordered by the year in which construction started, with 38 data fields per reef. The reef numbers 1 to 13 are those of the sections A.1 to A.13 of Appendix A. The table is compiled from the project dataset (files `reefs.json` and `reefs_table.csv`, held with the project files and available from the author). In the dataset every field is a record with a value, a unit, a status, the full citation and the place where the value was read (page, table, figure or file); only the status is reproduced here, and the principal sources and the methods behind each reef are given in Appendix A. The sources of the outcome and durability entries are cited in Table C4.

Key to the status markers. A marker follows each entry and applies to all numbers in the cell.

Marker	Meaning
s	**sourced**: a number or statement read in a cited source
d	**derived**: computed in this review from sourced inputs, or from a traced plan shape or a reconstructed 3D model; the method is in Appendix A
a	**assumed**: an assumption of this review, flagged in Appendix A (A-numbers)
?	**unknown**: not found in any source examined; no value is given

The outcome classes (works, works rarely, does not work, mixed or disputed, unknown) are this review's reading of the cited sources, marked ^{s} because the underlying statements are sourced; they are not measurements. Units: depths and heights in metres, depths positive downward; areas in m^2; volumes in m^3.

Table C1. Identification, construction, material and cost. Country codes: AUS Australia, NZ New Zealand, UK United Kingdom, USA United States. Price year marked * was not stated by the source and is assumed (the completion or construction year).

Reef	Lat., lon. (deg)	Built (completed)	Purpose	Material and size	Volume (m^3)	Cost, local currency (price year)	Cost (USD M, 2024 prices)
1. Burkitts Reef, Bargara (AUS)	-24.816, 152.467[d]	1997[s]	surf[s]	natural basalt boulders and platform, reshaped (at least 12 large boulders moved)[s]	300[s] (rock moved)	AUD 10,000[s] (1997)	0.015[d]
2. Cables Reef, Perth (AUS)	-32.016, 115.748[d]	1999[s]	surf[s]	granite boulders, 1.5 t body and 3.0 t perimeter stones, placed from a barge[s]	5,500[s] (rock)	AUD 1.8 M[s] (1999)	2.19[d]
3. Narrowneck Reef, Gold Coast (AUS)	-27.987, 153.434[d]	1999[s] (2000)	multi-purpose[s]	geotextile sand containers (about 20 m long, 3-4.5 m diameter, 160-300 t); about 400 initially, 534 placed by 2018[s]	70,000[s] (sand fill)	AUD 2.3 M[s] (1999)	2.79[d]
4. Pratte's Reef, El Segundo (USA)	33.921, -118.433[d]	2000[s] (2001)	surf[s]	sand-filled polypropylene geotextile bags, 3.0 x 2.1 x 1.2 m (110 in Phase I, +90 in Phase II)[s]	1,350[s] (sand, Phases I + II)	USD 550,000[s] (2000)	1.00[d]
5. Mount Maunganui Reef (NZ)	-37.645, 176.203[d]	2005[s] (2008)	multi-purpose[s]	24 sand-filled geotextile containers, up to 50 x 7.2 x 3.2 m[s]	2,800[s] (as built)	NZD 1.5 M[s] (2008*)	1.54[d]
6. Opunake Reef (NZ)	-39.459, 173.852[d]	2006[s] (never completed)	surf[s]	sand-filled geotextile bags on a thin sand layer over rock strata (27 bags in place April 2008)[s]	4,900[s] (design)	NZD 1.7 M[s] (2009)	1.55[d]
7. Boscombe Surf Reef (UK)	50.718, -1.839[d]	2008[s] (2009)	multi-purpose[s]	sand-filled non-woven geotextile containers, two layers (32-55 containers, sources differ; each up to 60-70 m long)[s]	12,800[s] (sand fill, designers)	GBP 3.2 M[s] (2009)	7.29[d]
8. Kovalam Reef (India)	8.387, 76.975[d]	2009[s] (2010)	multi-purpose[s]	sand-filled geotextile container strips, about 30 m long (5-6 strips fanning out)[s]	4,000[s] (basis not stated)	INR 75 M[s] (2010*)	2.36[d]
9. Borth reef (UK)	52.483, -4.056[d]	2011[s] (2012)	multi-purpose[s]	quarry rock armour (Types 1, 3, 4, 6; 0.005-10 t); about 51,000 t in the two mounds (30,119 m^3 x 1.7 t m^{-3}, derived)[s]	30,119[d] (rock, survey edge)	GBP 13 M[s] (2012*; whole Phase 1 scheme)	28.06[d]
10. Palm Beach Reef, Gold Coast (AUS)	-28.107, 153.471[d]	2019[s]	multi-purpose[s]	quarried basalt/greenstone boulders, about 60,000 t[s]	25,000[s] (rock, as stated)	AUD 18.2 M[s] (2019)	15.52[d]
11. Bunbury Airwave (AUS)	-33.328, 115.628[a]	2019[s] (installation failed)	surf[s]	inflatable Hypalon rubber bladder, 12 m diameter, ballasted with 140-150 t of beach sand[s]	105[d] (spherical-cap stand-in)	USD 400,000[s] (2019)	0.49[d]
12. Southern Ocean Surf Reef, Albany (AUS)	-35.018, 117.921[d]	2025[s]	surf[s]	quarried rock in three grades (belly, core, armour), 62,800 t in total[s]	34,000[s] (rock, design)	AUD 11.75 M[s] (2025*)	7.57[d]
13. Xala Reef, Jalisco (Mexico)	? (undisclosed)	2025[s] (completion not stated)	multi-purpose[s]	quarried boulders ("3,000 truckloads")[s]	?	?	?

Table C2. Plan shape and vertical geometry. All derived values come from the traced plan shapes and the reconstructed 3D models (Appendix A); the reef number is the same as in Table C1. Crest depth is the depth of the shallowest crest point below mean sea level (MSL) or lowest astronomical tide (LAT); a negative value is a crest above that datum. Seabed at toe is the depth of the bed (not of the reef surface) at the seaward edge of the traced outline on the crest transect. Depth gap = seabed at toe minus crest depth (MSL). Mean slope = depth gap divided by the horizontal ramp length from toe to crest. The dagger marks values referred to chart datum (LAT), not MSL (Opunake). Empty cells ("?") mean that no source gives the value and the reef could not be reconstructed (Burkitts, Xala), or that no crest or seabed depth is known (Kovalam).

Reef	Plan area (m^2)	Along × cross-shore (m)	Height (m)	Crest depth, MSL (m)	Crest depth, LAT (m)	Seabed at toe, MSL (m)	Depth gap (m)	Ramp length (m)	Mean slope (1:n)
1. Burkitts Reef	?	?	?	?	?	?	?	?	?
2. Cables Reef	4,049[d]	121 × 76[d]	?	1.55[s]	0.83[a]	6.51[d]	4.96[d]	41.5[d]	0.119 (1:8)[d]
3. Narrowneck Reef	6,239[d]	113 × 157[d]	2.2[d]	2.32[d]	1.44[d]	6.13[d]	3.81[d]	51.0[d]	0.075 (1:13)[d]
4. Pratte's Reef	435[d]	58.0 × 28.6[d]	1.4[d]	2.68[d]	1.21[d]	4.03[d]	1.35[d]	1.4[a]	1.000 (1:1)[a]
5. Mount Maunganui Reef	1,071[d]	67.9 × 60.4[d]	2.7[a]	2.03[d]	0.87[d]	4.73[a]	2.70[d]	5.1[d]	0.530 (1:2)[d]
6. Opunake Reef	2,970[s]	30 × 100[s]	3.5[s]	?	0.00[s]	3.00[s] †	3.00[d] †	?	?
7. Boscombe Surf Reef	4,042[d]	88.5 × 94.3[d]	3.2[d]	0.90[d]	-0.56[d]	5.46[d]	4.56[d]	14.0[d]	0.326 (1:3)[d]
8. Kovalam Reef	1,216[d]	59.0 × 41.0[d]	?	?	?	?	?	?	?
9. Borth reef	5,743[d]	196 × 140[d]	4.5[d]	-0.19[d]	-2.94[d]	4.26[d]	4.62[d]	38.0[d]	0.115 (1:9)[d]

Reef	Plan area (m^2)	Along × cross-shore (m)	Height (m)	Crest depth, MSL (m)	Crest depth, LAT (m)	Seabed at toe, MSL (m)	Depth gap (m)	Ramp length (m)	Mean slope (1:n)
10. Palm Beach Reef	11,972[d]	137 × 149[d]	3.9[d]	1.50[s]	0.62[d]	6.67[d]	5.17[d]	94.0[d]	0.055 (1:18)[d]
11. Bunbury Airwave	112[d]	12.1 × 12.1[d]	2.0[s]	0.68[d]	-0.02[d]	2.92[d]	2.24[d]	6.0[d]	0.373 (1:3)[d]
12. Southern Ocean Surf Reef	13,104[d]	127 × 153[d]	2.6[d]	1.00[s]	0.26[d]	7.33[d]	6.33[d]	20.0[d]	0.317 (1:3)[d]
13. Xala Reef	?	?	?	?	?	?	?	?	?

Table C3. Setting, typical conditions and outcome. Distance offshore is that of the crest from the shoreline definition used for each reef (see note 4 below). Arm angle is the angle between the reef arm (or the long axis of the outline) and the shoreline; two values are the two arms. For Narrowneck the value is the northern arm (the southern arm is 90.0 deg). For Pratte's Reef the angles hold if the V axis is shore-normal. Tide range: spring range (MHWS - MLWS: Narrowneck, Palm Beach, Mount Maunganui, Boscombe, Borth, Burkitts at Burnett Heads, Opunake at Port Taranaki) / mean range ((spring + neap)/2). At the diurnal and mixed sites the first value is not a spring range: it is the great diurnal range MHHW - MLLW at Pratte's (NOAA GT 1.654, MN 1.145 m) and the predicted annual range, about LAT to HAT, of the 2026 predictions of the Western Australia Department of Transport and Bureau of Meteorology at Cables (Fremantle, 1.14 m), Bunbury (1.20 m) and Albany (1.38 m) (Western Australia Department of Transport & Bureau of Meteorology, 2026); the Port Taranaki and Burnett Heads levels are from Land Information New Zealand (2026) and Maritime Safety Queensland (2026); these three are lower bounds (three of four quarterly blocks parsed) and micro-tidal in the sense of a range under 2 m. The Bunbury mean range (0.25 m, from a consultant's tidal planes) and the Cables mean range (0.80 m, NSW review, no definition) are probably too small and were not recomputed; the Kovalam entry carries no definition in its source. Typical swell is as stated in the source and is mostly undefined (Narrowneck 1.0 m, Mount Maunganui 1.5 m and 11 s, Boscombe 0.5 m and 4 s, Kovalam 1-2.5 m and 11-15 s: Blacka et al., 2013, Table 4.3; Burkitts is wave-face height; Opunake is the designed breaking-wave face height).

Reef	Crest distance offshore (m)	Arm angle to shoreline (deg)	Tide range, spring / mean (m)	Typical swell Hs (m)	Typical Tp (s)	Surf outcome	Surfable days per year
1. Burkitts Reef	?	?	2.33[d] / 1.74[d]	0.5-2 (wave face 2-6 ft)[s]	7-10[s]	works[s]	30[s]
2. Cables Reef	268[d]	27 / 60[d]	1.14[d] (annual) / 0.80[s]	1-2 (summer), 1.5-2.5 (winter)[s]	?	mixed/disputed[s]	142[s]
3. Narrowneck Reef	212[d]	72[d]	1.31[d] / 1.02[d]	1.0[s]	?	works rarely[s]	?
4. Pratte's Reef	103[d]	45 / 44[d]	1.65[d] (great diurnal) / 1.15[d]	?	?	does not work[s]	?
5. Mount Maunganui Reef	308[d]	50 / 55[d]	1.75[d] / 1.42[d]	1.5[s]	11[s]	works rarely[s]	?
6. Opunake Reef	300[s]	?	3.16[d] / 2.35[d]	1.8-2.5 (design wave face)[s]	?	does not work[s]	1-in-8 to 1-in-3 days (design aim)[s]
7. Boscombe Surf Reef	239[d]	55[d]	1.76[s] / 1.13[d]	0.5[s]	4[s]	works rarely[s]	77 (design baseline) to 154 (design target)[s]
8. Kovalam Reef	90[s]	38[d]	? / 1.00[s]	1-2.5[s]	11-15[s]	mixed/disputed[s]	?
9. Borth reef	330[d]	64[d]	4.30[d] / 3.00[d]	?	?	unknown[s]	?
10. Palm Beach Reef	243[d]	45[d]	1.31[d] / 1.02[d]	?	?	works[s]	?
11. Bunbury Airwave	38[d]	?	1.20[d] (annual) / 0.25[d]	?	?	does not work[s]	?
12. Southern Ocean Surf Reef	189[d]	71[d]	1.38[d] (annual) / ?	?	?	works[s]	146[d]
13. Xala Reef	?	?	? / ?	?	?	unknown[s]	?

Table C4. Durability and shoreline or sediment outcome as reported. The entries condense the dataset fields `durability` and `sediment_shoreline_outcome`; each carries its source. A question mark means that nothing was found.

Reef	Durability	Shoreline and sediment outcome
1. Burkitts Reef	No reported maintenance, storm damage or rework in nearly three decades (status January 2024); still surfed (Howells & O'Neal, 2024).	? Not assessed in any source found.
2. Cables Reef	No structural damage reported. Surveyed rock volume 4,671 m^3 (November 1999) and 5,219 m^3 (February 2002) differ by less than the survey error (this review, from Department of Transport WA soundings); no information on long-term stability (Blacka et al., 2013, p. 93).	None relevant: not designed for coastal protection; stable limestone shoreline, negligible longshore transport (Blacka et al., 2013, p. 67 and p. 92).
3. Narrowneck Reef	Containers stable in waves up to 8-10 m; settlement of 1-2 m locally; damage by anchors, spears, knives and boards; renewal in 2017-18 with 84 containers (Blacka et al., 2013, pp. 97-98; Corbett et al., 2023).	Mixed or disputed. A published 20-50 m salient (Mead & Borrero, 2011, p. 4) is read by Blacka et al. (2013, pp. 98-100) as a probable misreading of an arbitrary dune reference line; sand increases are mostly in the offshore bars (Blacka et al., 2013; Nettle, 2019).

Reef	Durability	Shoreline and sediment outcome
4. Pratte's Reef	Bags moved, sank and were buried; the polypropylene largely disintegrated within about two years; crest dropped 1.5 m in the year after Phase II; removed in 2008 and 2010 (Blacka et al., 2013, pp. 116-119; Borrero & Nelsen, 2003; Fontaine, 2008).	No discernible effect on bathymetry or beach in monthly profiles and quarterly surveys (Blacka et al., 2013, p. 117).
5. Mount Maunganui Reef	Containers ruptured and lost sand during filling; one of the largest was damaged beyond repair; apex bags deflated to -2.0 to -2.4 m CD by 2013; reef removed 25 September to November 2014 (Mead & Borrero, 2011, p. 3; Dahm & Gibberd, 2013).	Weak or disputed. About 25 m of beach widening and an influence of about 150 m to the south-east (Mead & Borrero, 2011, p. 5), read by Blacka et al. (2013, p. 103) as subtle, with protection rated "Negligible".
6. Opunake Reef	Structure began to break apart and pieces washed ashore; designers called bag failure the "fatal flaw" of the technology (Reid & Sumner, 2021); 27 bags in place in April 2008; no formal removal reported.	? Not found.
7. Boscombe Surf Reef	Containers failed from March-April 2011; reef closed 31 March 2011; repairs aborted; operator liquidated in 2012; structure still present (Rendle & Davidson, 2012; this review, Appendix A.7).	Designers: beach in the lee accreted by up to 60 m within six months (Mead & Borrero, 2011, p. 5); later monitoring found no noticeable long-term effect on the 0 m contour (Rendle & Davidson, 2012); protection rated "Minor" (Blacka et al., 2013).
8. Kovalam Reef	Containers torn, deflated and sunk; damage hampering the protective effect by 2013 (E. Rendle, pers. comm., 2013, in Blacka et al., 2013, p. 70); degraded in place, no formal removal (Raised Water Research, n.d.-c).	Designers: beach 10-15 m wider in the monsoon and up to 30 m in the non-monsoon season; fewer than three years of monitoring; independent assessment pending (Mead & Borrero, 2011; Blacka et al., 2013, p. 70).
9. Borth reef	No damage, deflation or removal found (imagery 2012-2024, LiDAR 2022); rock armour sized with three-dimensional physical modelling; defences withstood the December 2011 storms (Hansford, 2011; Daily Post, 2013).	Flood protection reported effective from December 2011; the scheme combines reef, groynes, breakwater and 84,500 t of shingle nourishment; no quantified shoreline response (Blacka et al., 2013, p. 70: performance "yet to be determined").
10. Palm Beach Reef	Stable, with no significant movement or settlement (Hunt et al., 2022, monitoring abstract); no damage or removal found.	About 550,000 m^3 of sand retained over the first two years (ABC News, 2022); a journalistic statement, not a monitoring report.
11. Bunbury Airwave	Bladder torn along a glued seam during installation (December 2019); removed within about three days (ABC News, 2019).	? Not applicable (never operational).
12. Southern Ocean Surf Reef	Built with graded rock; a storm on 26 May 2025 moved some core rock; about 18 high rocks repositioned in February 2026 after user injuries; no other damage reported (Sheil, 2026).	? Not found.
13. Xala Reef	? Unknown (announced March-April 2026).	? Unknown.

Table C5. Number of dataset fields (of 38) in each status class. The fields counted include the identification fields (name, country, coordinates), so the totals show how much of the record is measured, derived, assumed or missing. s = sourced, d = derived, a = assumed, ? = unknown.

Reef	s	d	a	?	Fields
1. Burkitts Reef	16	6	0	16	38
2. Cables Reef	20	15	1	2	38
3. Narrowneck Reef	16	19	0	3	38
4. Pratte's Reef	15	18	2	3	38
5. Mount Maunganui Reef	16	16	3	3	38
6. Opunake Reef	21	7	1	9	38
7. Boscombe Surf Reef	19	19	0	0	38
8. Kovalam Reef	18	8	0	12	38
9. Borth reef	13	20	0	5	38
10. Palm Beach Reef	16	18	0	4	38
11. Bunbury Airwave	12	16	2	8	38
12. Southern Ocean Surf Reef	14	18	0	6	38
13. Xala Reef	10	0	0	28	38

Notes to Tables C1-C5

1. **Datums.** Each 3D model is referenced to mean sea level (z = 0 at MSL). Conversions applied: Narrowneck and Palm Beach $z_{MSL} = z_{AHD}$ - 0.12 m (Maritime Safety Queensland, Gold Coast Seaway; the rule used in the papers would shift this by 0.24 m); Cables, Bunbury and Albany: Australian Height Datum (AHD) taken as MSL (Cables low water datum is 0.716 m below AHD; Albany LAT is 0.74 m below AHD); Mount Maunganui $z_{MSL} = z_{CD}$ - 1.13 m (Land Information New Zealand, Tauranga, present MSL); Boscombe $z_{MSL} = z_{ACD}$ - 1.40 m (survey zero assumed to be chart datum); Borth $z_{MSL} = z_{ODN}$ - 0.31 m; Pratte's $z_{MSL} = z_{MLLW}$ - 0.849 m. Opunake and Kovalam have no usable datum.
2. **Ramp and slope.** The ramp is the horizontal toe-to-crest distance on the shore-normal transect through the shallowest crest cell; the crest edge is the seaward-most cell within 0.25 m of that column's maximum (the dataset gives the sensitivity to 0.10 and 0.50 m). Ramp and slope are derived from the reconstructed 3D models for Narrowneck, Cables, Albany, Palm Beach and Borth

(median over the columns of the long diagonal mound), from the idealised loft of the 3D model for Boscombe (the 1:3 flank is an assumption), by hand from the 2013 digital elevation model for Mount Maunganui, and from the spherical-cap stand-in for Bunbury. **Pratte's ramp and slope are assumed (1:1 bag face).** Where a sourced crest depth exists (Cables, Palm Beach, Albany) the depth gap uses it. Only Narrowneck, Cables, Albany, Palm Beach, Borth and Boscombe have a ramp from audited geometry; relations should not be fitted through the whole set without the status markers.

3. **Plan areas are definitions, not always the reef.** Narrowneck: visible container field on the aerial of 8 August 2020, a lower bound. Cables: original design plan. Pratte's: Phase I design layout. Mount Maunganui: the -2.0 m CD contour of the 2013 multibeam survey (the 2008 toe outline is 1,424 m^2). Opunake: design bounding rectangle 99 x 30 m, not the traced reef; the bag layout is not published, and the along-shore and cross-shore values in Table C2 are the published design size with unknown orientation. Boscombe: outline traced on the image of 28 September 2011. Kovalam: photo trace at an assumed camera height of 60 m (550-1,700 m^2 for 30-80 m), low confidence. Borth: **area is the visible rock edge on the photograph of 17 September 2024; the volume (30,119 m^3) belongs to the laser-survey edge of 2022 (8,675 m^2)**, the default of the 3D model. Palm Beach: toe outline. Albany: Bluecoast design outline (the dark rock patch visible in 2026 is 10,918 m^2). Bunbury: circle of 12 m diameter traced on a drone photograph. Narrowneck volume (70,000 m^3) is the sand fill stated by the designers; the 3D mound envelope is a different quantity.
4. **Distance offshore** is measured from different shoreline definitions: the MSL waterline of the image used (Narrowneck 8 August 2020, Palm Beach 1 December 2025, Pratte's), the wet/dry-sand edge (Mount Maunganui, Albany beach-face line), the back-of-beach defence line (Borth), the pre-reef shoreline (Kovalam, as stated in the NSW review) and a survey surf line (Boscombe). Differences below 30 m are noise.
5. **Costs.** Costs are in nominal local currency as stated by the source, for what each source calls the reef; they may include studies. The Borth figure (GBP 13 M) is for the whole Phase 1 defence scheme and is not a reef cost. USD = local amount divided by the World Bank annual-average official exchange rate of the price year; USD (2024) = nominal USD multiplied by the US consumer price index of 2024 divided by that of the price year (2025 uses the 2024 index) (World Bank, 2026). Both are derived and inherit the uncertainty of the price year.
6. **Years.** "Built" is the start of construction and "completed" the end of the main phase. Reefs with soft structures that failed or were removed are listed with their original construction years.
7. **Coordinates** are the centres of the traced outlines (derived); Bunbury is an estimated centre (assumed); Pratte's is the centre of the survey window of Borrero and Nelsen (2003) (position +-150 m); Opunake is the centre of the consent area; Burkitts is the platform centre; Xala is undisclosed.
8. **Not tabulated** (available in the dataset): seabed depth at the crest, mean tide range definitions, design-wave statements, swell direction, arm angle to the swell, and the alternative values (other published or derived values for the same quantity, each with its own status and source).
9. **Cautions.** Typical swell (Hs, Tp, direction) is unknown for most reefs because the design reports of the original designers were not read in full; the Haifa wave data are not part of this table. Cost conversions assume the price year given. Values of Mount Maunganui (assumed bed), Bunbury (stand-in surface) and Pratte's (assumed slope) are weak and are marked.

Appendix D. Other reefs and geotextile structures named by Bar and Drimer (2023) and Israeli geotube projects

This appendix lists the structures that are not among the 13 surf reefs of the survey (Section 3, Appendix C) but are named in the survey paragraph of Bar and Drimer (2023, p. 2) or are relevant to the Israeli coast. None of them was built as a surf reef. Of the eleven projects and studies named in Bar and Drimer (2023), three (Narrowneck, Pratte's Reef and Mount Maunganui) are among our 13 reefs; the other eight are D.1 to D.8: four (D.1 to D.4) are geotextile-tube (Geotube) shore-protection breakwaters, one (D.8) is a cliff-foot geotube scheme, one (D.5) is a nature-based breakwater, and two (D.6, D.7) are design studies; D.9 to D.11 are further Israeli geotube projects. Each entry gives type, year, place, purpose, material and size, outcome, and the verification level of this review. The levels are: READ (the cited text was opened and the number checked), ABSTRACT (only an abstract or catalogue record was reached), and VENDOR (a manufacturer page, treated as promotional). Nothing is stated here that is not in the cited source; where the source is silent the entry says "not found". All sources were read or checked on 15 October 2026. Page numbers of conference papers and of the Israeli audit reports are PDF pages.

Bar and Drimer (2023) is not a survey. Their review of built projects is one paragraph; the remainder of the paper develops and validates a two-dimensional boundary-element wave-flume model for geotextile tubes as submerged impermeable structures. They borrow the statement that "most of the projects have failed" from the Raised Water Research website, without a list or a definition of failure (Bar & Drimer, 2023, p. 2). Of the eight built projects they name, only three (Narrowneck, Pratte's Reef and Mount Maunganui) are among our 13 reefs; ten of our reefs are not mentioned by them.

Table D1. Structures named by Bar and Drimer (2023) that are not in the 13-reef survey, and Israeli geotube projects.

Entry	Place	Type	Built	Outcome reported	Level
D.1 Young-Jin	Gangneung, Korea	submerged geotube detached breakwaters	about 2004-05 (d)	shoreline +2.4 to 7.6 m in 12 months; local re-erosion	READ
D.2 Yucatan	Progreso area, Mexico	submerged geotube breakwaters	2005-07	mixed; downdrift shadow effect	READ
D.3 Al Aqah	Fujairah, UAE	geotube groins and submerged breakwater	2010-14	“well controlled” (manufacturer)	VENDOR
D.4 Sigandu	Batang, Java, Indonesia	low-crested geotube breakwaters	from Sep 2014	salients and tombolos within a month	ABSTRACT
D.5 Living Breakwaters	Staten Island, USA	armour-stone and eco-concrete breakwaters	2021-24	too recent	READ (press)
D.6 Portuguese MFAR	Portugal	multifunctional reef, design study	not built (as far as found)	none	bibliographic
D.7 Babilonia MFAR	Guardamar, Spain	multifunctional reef, design study	not built	none	ABSTRACT
D.8 Ashkelon North	Ashkelon, Israel	cliff-foot geotubes (48 tubes, 5 clusters)	2017/18-2019	several tubes torn in first winter; later collapsed	READ
D.9 Ashkelon South	Ashkelon, Israel	armoured geotube pilot	from Sep 2023	pilot, not built as a structure	READ
D.10 Hof HaTzuk	Tel Aviv, Israel	submerged geotube breakwater	planned 2019, stopped 2022	not built	READ/secondary
D.11 Ashkelon trial	Ashkelon, Israel	geotube trial, about 20 m tubes	2005-06	storm damage and vandalism	secondary

D.1 Young-Jin beach, Gangneung, Gangwon-do, east coast of Korea

Type: submerged double-lined geotextile-tube detached breakwaters for shore protection; an example of submerged geotube shore protection, not a surf reef. *Year:* not stated; the paper (2006) already reports 12 months of monitoring, so construction was about 2004-2005 (derived; assumption). *Purpose:* severe erosion threatened scenery and public property; a tetrapod breakwater would have needed heavy equipment, so geotubes were chosen as a cheaper and faster alternative (Oh & Shin, 2006a, pp. 1-2). *Material and size:* woven geotextile of at least 700 g/m^2; sand fill with D_{10} = 0.50 mm; tubes filled by cutter-suction dredge (about 500 m^3/h). Double-lined tubes in about 3.0 m of water, 90-100 m from the shoreline, each tube 50 m long and 1.8 m in diameter, four twin units at 20.0 m spacing covering 240 m of shoreline; model tests led to a design with the crest at the water surface (Oh & Shin, 2006a, pp. 2-4). *Outcome:* the shoreline advanced about 2.4-7.6 m seaward within 12 months and sand accumulated around the tubes; at one benchmark the shoreline advanced 5.3 m in three months and then re-eroded, which the authors attribute to changed tidal flow; seaweed colonised the tubes after one year. The authors state that long-term effectiveness cannot be judged yet because of severe storms (Oh & Shin, 2006a, p. 4). *Cost and contractor:* not stated. *Relevance:* a zero-freeboard soft structure at about 3 m depth with local re-erosion and no durability record beyond 12 months. *Citations:* Oh and Shin (2006a), read; the journal version (Oh & Shin, 2006b) is paywalled and was not read. *Level:* READ (conference paper).

D.2 Yucatan coast (Progreso area), Mexico

Type: geotextile tubes as low-crested or semi-submerged breakwaters for beach restoration. *Year:* field installation 2005 (4 km of tubes by September 2005), further sites 2005-2007 (Alvarez et al., 2006; González Leija et al., 2014). The paper that Bar and Drimer (2023) cite is a laboratory flume study, not the field project. *Purpose:* shoreline retreat of about 1 m/yr and more over 15 years; federal beach rehabilitation programme from 2001 (Alvarez et al., 2006). *Material and design inputs:* woven polypropylene tubes with a scour apron; breaking wave height below 1.0 m (93 % of the time), T_z 4-8 s, bed slope 0.1-0.5 %, tide range 0.90 m, littoral drift 60,000 m^3/yr, D_{50} = 0.30 mm; the designers computed the Iribarren number, found spilling breakers, and set the crest so that high-tide freeboard avoids reflection (Alvarez et al., 2006). *Outcome:* mixed. “In some cases shore protection and beach stabilization was achieved … but in other places the results have been far from the expected”, with a downdrift shadow effect; tubes were set at about -0.3 m relative to MSL (González Leija et al., 2014). The laboratory study found that the transmission coefficient depends on the ratio of submergence depth to incident wave height, and that tubes crowned well below mean water level make waves “steepen and break more violently”. *Cost and contractor:* not found. *Citations:* Alvarez et al. (2006), González Leija et al. (2014) read; Alvarez et al. (2007) not read. *Level:* READ (magazine article and one-page poster).

D.3 Al Aqah Beach, Fujairah, UAE

Type: geotextile-tube groins plus a seaward submerged breakwater, for beach nourishment and erosion control; not a surf reef. *Year:* after cyclone Gonu (2007); manufacturer’s case-study timeline 2010-2014. *Purpose:* erosion over “several kilometers”; rip-rap groins had sunk under repeated wave action (ACE Geosynthetics, n.d.). *Material and size:* geotextile tubes of 8.6-17.2 m circumference and 52-77 m length, two arms of 200 m each, filled with sand in situ. Crest depth, fill height and volume: not found. *Outcome:* the manufacturer states that erosion is “well-controlled”; no independent monitoring found. *Cost:* not found. *Citations:* ACE Geosynthetics (n.d.), read; Chien et al. (2013) was not accessible (abstract level only); Chien et al. (2014) as reported by Bar and Drimer (2023), full text not accessed. *Level:* VENDOR.

D.4 Sigandu Beach, Batang Regency, Central Java, Indonesia

Type: low-crested geotextile-tube breakwaters for shore protection. *Year:* installation from September 2014; paper 2015. *Purpose:* an eroding tourist beach, attributed to the jetty of the Batang commercial port; the beach receded about 5 m in 8 years (Solmax, 2023). *Material and size (vendor):* tubes of 9.5 m circumference, design filled height 1.5 m, length 25 m, on a scour apron with two pilot tubes; number of tubes, crest depth and offshore distance not found. *Outcome:* about one month after installation, rapid sedimentation formed salients and tombolos behind the structures (Sulaiman et al., 2015, abstract). A later paper by the same group (Sulaiman et al., 2017) may report the longer-term outcome; it was not read. *Citations:* Sulaiman et al. (2015) abstract only (full text returned HTTP 403 and was not bypassed); Solmax (2023) read. *Level:* ABSTRACT and VENDOR.

D.5 Living Breakwaters, Tottenville, Staten Island, New York

Type: eight partially submerged breakwaters of armour stone and ecologically enhanced concrete units, with “reef ridges” and “reef streets” for habitat, plus sand replenishment; coastal risk reduction and ecology, no surfing. *Year:* construction from 2021; completion announced in September 2024. *Size and cost:* 2,400 linear feet of breakwaters, USD 111 million, designed to reduce the waves that reach buildings and roads to below three feet (about 0.9 m) (Hochul, 2024). *Outcome:* too recent for performance data. *Design papers:* Baker et al. (2018), abstract only; the conference paper that Bar and Drimer (2023) cite for this project was not accessed. *Relevance:* a rock-plus-eco-concrete submerged breakwater, comparable to the ECOncrete applications in Israel. *Level:* READ (press release).

D.6 Portuguese multifunctional artificial reef (design study)

A preliminary design of a multifunctional reef (coastal protection and surfing) for the Portuguese coast (ten Voorde et al., 2009), with a follow-up on the influence of the parameters in the vertical plane (Antunes do Carmo et al., 2011). Neither paper was read and no evidence that the reef was built was found. *Level:* bibliographic only.

D.7 Babilonia beach, Guardamar del Segura, Alicante, Spain (design study)

A design method for multifunctional artificial reefs applied to the eroding Babilonia beach on the Spanish Mediterranean coast (López et al., 2016). The abstract, as relayed by the University of Alicante record, states that the reef would give a surfable wave “60 % of the time” for surfers of low to intermediate skill. This figure was not confirmed in the full text (the open-access download returned HTTP 401 and was not bypassed) and must not be quoted before it is. The paper is the closest published analogue to a microtidal Mediterranean surf reef with a surfer-skill statement. *Level:* ABSTRACT.

D.8 Ashkelon North geotubes, Israel

Type: submerged geotextile-tube structure of 48 tubes in five clusters at the foot of the eroding Ashkelon cliff; urgent, temporary marine protection until permanent breakwaters are approved. It was the first geotube project in Israel (Bar & Drimer, 2023, p. 2, citing a 2016 design report by Gouaud that could not be located online; not verified). *Dates:* the State Comptroller of Israel (2020, PDF pp. 71, 83) gives July 2017 as the contractor’s start and September 2019 as completion; the Government Company for the Protection of Mediterranean Sea Cliffs (MCCP, 2026) lists April 2018 to April 2019; Rinat (2019) says the tubes were laid in summer 2018. *Size:* 48 tubes (State Comptroller of Israel, 2020); 50 according to Rinat (2019); tube dimensions are not stated in any source read. The figures of 10 dunam and 4 m height in the author’s notes were not found in any source. *Cost:* about NIS 11 million (State Comptroller of Israel, 2020); NIS 10 million (Rinat, 2019); NIS 10.5 million in the author’s notes. *Outcome:* “several of the geotubes were damaged and torn shortly after being laid in the sea, after the winter storms, and had to be repaired” (State Comptroller of Israel, 2020, p. 71). In early February 2019 the contractor reported severe damage of unknown cause and halted laying; the last storm “was not so severe” (Rinat, 2019). The number of destroyed tubes is not stated in any source read; the count “15 of 48” in the author’s notes and the count “17 by storms, 2 by boats” in an earlier repository dossier are unsupported. During maintenance in late 2021, 12 geotubes were replaced (State Comptroller of Israel, 2024, p. 48). The 2024 report states that the geotube marine protection in Ashkelon North “collapsed” (p. 29) and that geotubes were considered “less effective” there, with five rock breakwaters planned (p. 53). The same report lists general geotube threats: poor design assumptions, poor filling, and damage by propeller blades, anchors and divers’ knives (p. 9). *Surfers and ecology:* the local surfers’ organisation opposed the tubes (Rinat, 2019). No monitoring results were found. *Netanya:* the author recalled torn tubes at Netanya in winter 2019; no source shows a geotube built or contracted at Netanya, whose cliff protection is rock breakwaters (State Comptroller of Israel, 2020, p. 71; 2024, p. 48; MCCP, 2026). The timing the author remembers (first winter after laying) matches Ashkelon North, so Ashkelon is the site used in this review. *Level:* READ (two official reports and one press article).

D.9 Ashkelon South (area 39), Israel

Geotubes were not built. A December 2017 committee decision limited stage 1 to sand feeding; from September 2023 the company has piloted an “armoured geotube” (fabric with metal conductors for low-voltage mineral precipitation), designed in response to “the low durability and survivability of the fabric sleeves used at Ashkelon North” (State Comptroller of Israel, 2024, pp. 54, 55, 57). *Level:* READ.

D.10 Hof HaTzuk (“Cliff Beach”), Tel Aviv

A submerged geotube breakwater was planned (tender 2019), opposed by surfers and an environmental group, and stopped at tender stage in 2022 on the Ministry’s recommendation; it was not built (State Comptroller of Israel, 2024, p. 11; the 2019 press reports were not re-opened). *Level:* READ for the 2024 report, secondary for the rest.

D.11 Ashkelon geotube trial, 2005-2006

An earlier trial by the Ashkelon Economic Company (about NIS 1.7 million, tubes of about 20 m) suffered storm damage and vandalism according to a Haaretz report of 6 April 2006 as recorded in an earlier dossier of this review; the report was not re-opened and the entry is not verified. *Level:* secondary.

D.12 Types named without a site

Bar and Drimer (2023, p. 1) mention floating breakwaters, Bragg breakwaters, submerged reef balls and an artificial mangrove-root system as alternatives that “may present potential advantages in cost, hydrodynamics and environmental features”. No project data are given and none of the underlying papers was read.